

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

AMPHASTAR PHARMACEUTICALS, INC.,
Petitioner,

v.

AEGIS THERAPEUTICS, LLC,
Patent Owner.

IPR2021-01324
Patent 10,682,414 B2

Before ROBERT A. POLLOCK, TIMOTHY G. MAJORS, and
CYNTHIA M. HARDMAN, *Administrative Patent Judges*.

HARDMAN, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining Some Challenged Claims Unpatentable
35 U.S.C. § 318(a)
Denying Petitioner's Motion to Exclude Evidence
37 C.F.R. § 42.64

I. INTRODUCTION

This is a Final Written Decision in an *inter partes* review challenging the patentability of claims 1–20 of U.S. Patent No. 10,682,414 B2 (“the ’414 patent,” Ex. 1001). We have jurisdiction under 35 U.S.C. § 6.

Petitioner has the burden of proving unpatentability of the challenged claims by a preponderance of the evidence. 35 U.S.C. § 316(e) (2018). Having reviewed the parties’ arguments and cited evidence, for the reasons discussed below, we find that Petitioner has demonstrated by a preponderance of the evidence that claims 1, 2, and 7–17 are unpatentable, but has not demonstrated that claims 3–6 and 18–20 are unpatentable. Additionally, for the reasons discussed below, we deny Petitioner’s Motion to Exclude (Paper 37).

A. *Procedural History*

Petitioner Amphastar Pharmaceuticals, Inc. filed a Petition for *inter partes* review of claims 1–20 of the ’414 patent. Paper 1 (“Pet.”). Patent Owner Aegis Therapeutics, LLC filed a Preliminary Response to the Petition. Paper 7. The parties further submitted an authorized Reply and Sur-reply to the Preliminary Response. Papers 10, 11. In view of the then-available preliminary record, we instituted an *inter partes* review. Paper 12 (“Dec.”).

After institution, Patent Owner filed a Response. Paper 20 (“PO Resp.”). Petitioner filed a Reply. Paper 25 (“Reply”). Patent Owner filed a Sur-reply. Paper 35 (“Sur-reply”).

Petitioner filed a Motion to Exclude Evidence. Paper 37. Patent Owner opposed that motion. Paper 39. Petitioner filed a Reply in support of the motion. Paper 40.

On November 16, 2022, we held an oral hearing, the transcript of which is of record. Paper 42 (“Tr.”).

B. Real Parties in Interest

Petitioner identifies itself as the real party in interest (“RPI”). Pet. 1. Patent Owner identifies itself and ARS Pharmaceuticals, Inc. as RPIs. Paper 5, 1.

C. Related Matters

Petitioner identifies U.S. Patent Application No. 16/869,461 (filed May 7, 2020) as related to the ’414 patent. Pet. 1. Patent Owner identifies no related matters. Paper 5, 1.

D. The ’414 Patent (Ex. 1001)

The ’414 patent, titled “Intranasal Epinephrine Formulations and Methods for the Treatment of Disease,” relates to intranasal epinephrine formulations and methods of treating anaphylaxis.¹ Ex. 1001, code (54), 1:17–19. Per the Specification, “[a]naphylaxis is a severe, potentially life-threatening type-1 hypersensitivity reaction (systemic allergic reaction) that affects many body systems.” *Id.* at 1:38–40. The Specification notes that although epinephrine injection is the standard treatment for anaphylaxis,

¹ The ’414 patent issued on an application filed May 22, 2019, which application is a continuation of an application filed February 6, 2019, and further claims priority to a provisional application filed December 21, 2018. Ex. 1001, codes (22), (60), (63). Petitioner assumes that the challenged claims are entitled to claim priority to December 21, 2018. Pet. 8–9. Patent Owner notes the priority claim to the December 21, 2018 provisional application, but does not appear to take a position on priority date. *See* PO Resp. 6. For purposes of this Decision, we will consider December 21, 2018, as the priority date but note that our analysis on this record would not change if the filing dates in February or May 2019 were used.

there have been “numerous reports of the highly variable exposures from injection products depending on the location of the injection (intramuscular or subcutaneous), and other factors such as body mass index (BMI).” *Id.* at 1:60–2:9. The Specification identifies a “need for alternative, needle-free and non-invasive methods for dosing epinephrine.” *Id.* at 2:24–25.

Challenged claim 1 relates to a method of treating a type-1 hypersensitivity reaction in a mammal via intranasal administration of an aqueous nasal spray pharmaceutical formulation that includes between about 0.1 mg and about 2.4 mg of epinephrine or a salt thereof. *Id.* at 72:15–23.

The Specification includes exemplary formulations (*id.* at 60:63–62:21, Tables 2–4) and two clinical studies (*id.* at 63:30–71:58). The first clinical study assesses the comparative bioavailability of epinephrine after intranasal administration and intramuscular administration. *Id.* at 63:30–39. The intranasal formulation comprised 0.3 mg epinephrine, dodecylmaltoside (“DDM”²), benzalkonium chloride (“BKC”³), and ethylenediaminetetraacetic acid (“EDTA”). *Id.* at 64:26–31. The intramuscular injection delivered 0.3 mg of epinephrine. *Id.* at 64:31–33. The Specification reports that “[m]ean plasma concentration of epinephrine from IN [intranasal] administration remained significantly below that of

² DDM, an alkylglycoside (or alkylsaccharide), functions as an absorption enhancer, and is sold under the tradename INTRAVAIL. Ex. 1001, 35:50–36:14, 46:67–47:11. The INTRAVAIL tradename covers a class of alkylsaccharides. Ex. 1017, 6–7.

³ Benzalkonium chloride is also sometimes abbreviated “BZK.” *See, e.g.*, Ex. 1001, 37:33; PO Resp. 7–8.

epinephrine from intramuscular epinephrine.” *Id.* at 66:15–18; *see also id.* at Fig. 3.

The second clinical study first assesses “the optimal dose of a formulation of intranasal epinephrine,” then compares the bioavailability of that optimal intranasal dose to an intramuscular dose. *Id.* at 66:49–57. Subjects first received escalating intranasal doses of 0.5 mg, 1.0 mg, and 2.0 mg epinephrine, where the formulations again comprised epinephrine, DDM, BKC, and EDTA. *Id.* at 66:62–67:33. The 1.0 mg and 2.0 mg intranasal formulations exhibited a lower $T_{\max}$ than the intramuscular formulation exhibited in the first clinical study, and the area under the plasma concentration time curve (“AUC”) was higher for all intranasal formulations than for the intramuscular formulation. *Id.* at 69:39–58, Figs. 4–7.

The bioavailability of the 1.0 mg intranasal epinephrine dose was then compared to that of a 0.3 mg epinephrine intramuscular injection. *Id.* at 66:62–67:17. According to the Specification, “the plasma time vs. concentration curve for 1.0 mg IN epinephrine is very similar to that for 0.3 mg IM [intramuscular] epinephrine (EpiPen®) administered in the thigh.” *Id.* at 69:66–70:42, Figs. 8, 9. Per the Specification, these data “demonstrate that 1.0 mg intranasal epinephrine can be formulated to be highly similar to or better than a 0.3 mg intramuscular injection of epinephrine.” *Id.* at 45–48.

E. The Challenged Claims

Petitioner challenges all claims (1–20) of the ’414 patent. Pet. 1.

Claim 1, the only challenged independent claim, reads:

1. A method of treatment of a type-1 hypersensitivity reaction in a mammal comprising administering intranasally to the mammal in need thereof an aqueous nasal spray pharmaceutical formulation; wherein the aqueous nasal spray pharmaceutical formulation comprises:
 - an anaphylaxis active ingredient, wherein the anaphylaxis active ingredient consists of between about 0.1 mg and about 2.4 mg of epinephrine, or a salt thereof in a single dose.

Ex. 1001, 72:15–23. The challenged dependent claims add additional features. For example, claim 3 depends from claim 1 and requires that the intranasal administration of the claimed nasal spray “provides a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis.” *Id.* at 72:33–37. Claim 7 depends from claim 1 and requires, among other things, that the claimed nasal spray comprises “one or more absorption enhancement agents.” *Id.* at 72:54–62. Claim 11 depends from claim 7 and requires that BKC is present in specific amounts, as the sole absorption enhancement agent or with one or more other absorption enhancement agents. *Id.* at 73:31–43. Claims 19 and 20 depend (directly or indirectly) from claim 3, and require that the nasal spray comprises, among other things, DDM and BKC. *Id.* at 74:23–45.

F. The Asserted Grounds of Unpatentability

We instituted trial based on the following three grounds of unpatentability:

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1–18	§ 102	Potta ⁴
1–20	§ 103	Potta, Maggio, ⁵ Munjal ⁶
1–20	§ 103	Srisawat, ⁷ Maggio, Munjal

Dec. 45; Pet. 4.

Petitioner support its contentions with two declarations from Hugh D. C. Smyth, Ph.D. (Exs. 1002, 1083), a declaration from Steven F. Weinstein, M.D. (Ex. 1003), and a declaration from S. Craig Dyar, Ph.D., R.Ph. (Ex. 1082). Patent Owner support its contentions with a declaration from Michael A. Kaliner, M.D. (Ex. 2013) and a declaration from Joseph W. Wyse, Ph.D. (Ex. 2014).

⁴ Potta et al., U.S. Patent Pub. 2018/0289617 A1, published October 11, 2018 (“Potta,” Ex. 1004).

⁵ Maggio, U.S. Patent Pub. 2014/0107145 A1, published April 17, 2014 (“Maggio,” Ex. 1005). The parties sometimes refer to this reference as “Maggio ’145.”

⁶ Srisawat et al., *A Preliminary Study of Intranasal Epinephrine Administration as a Potential Route for Anaphylaxis Treatment*, 34 ASIAN PAC. J. ALLERGY. IMMUNOL. 38 (2016) (“Srisawat,” Ex. 1006).

⁷ Munjal et al., *A Randomized Trial Comparing the Pharmacokinetics, Safety, and Tolerability of DFN-02, an Intranasal Sumatriptan Spray Containing a Permeation Enhancer, with Intranasal and Subcutaneous Sumatriptan in Healthy Adults*, 56 HEADACHE 1455 (2016) (“Munjal,” Ex. 1020).

II. ANALYSIS

A. *Level of Ordinary Skill in the Art*

We consider the grounds of unpatentability in view of the understanding of a person of ordinary skill in the art (sometimes referred to herein as “POSA” or “skilled artisan”) as of December 21, 2018. *See supra*

n.1. Petitioner contends:

a person of ordinary skill in the art (POSA) would have been composed of an individual or a team that would have included (1) an advanced degree (e.g., M.D., Pharm. D., Ph.D., or D.V.M.) and training in pharmacology, pharmaceutical sciences or related discipline or experience treating type-1 hypersensitivity reactions in a mammal and (2) at least two years of experience with the development of small molecule pharmaceutical formulations and the development and science of intranasal or mucosal delivery systems or the treatment of type-1 hypersensitivity reactions in a mammal.

Alternatively, the POSA would have been composed of an individual or a team that would have included (1) a bachelor’s degree or similar; and (2) more than seven years of experience with the development of small molecule pharmaceutical formulations and the development and science of intranasal or mucosal delivery systems or the treatment of type-1 hypersensitivity reactions in a mammal.

To the extent necessary, a POSA may collaborate with one or more other POSAs, such as an expert in pharmaceutical formulations or a medical doctor, for one or more aspects with which the other POSA may have expertise, experience, and/or knowledge.

Pet. 7–8 (spacing added). For purposes of this proceeding, Patent Owner accepts Petitioner’s proposal. PO Resp. 9.

We adopt Petitioner’s proposed level of ordinary skill in the art because it is unopposed and consistent with the cited prior art.

Based on their statements of qualifications and curricula vitae, we find Petitioner’s declarants (Drs. Smyth, Weinstein, and Dyar), and Patent Owner’s declarants (Drs. Kaliner and Wyse) qualified to provide technical opinions from the perspective of a person of ordinary skill in the art. Ex. 1002 ¶¶ 2–12, pp. 292–341 (Dr. Smyth’s statement of qualifications and curriculum vitae); Ex. 1003 ¶¶ 2–12, pp. 208–17 (Dr. Weinstein’s statement of qualifications and curriculum vitae); Ex. 1082 ¶¶ 2–18, pp. 63–65 (Dr. Dyar’s statement of qualifications and curriculum vitae); Ex. 2013 ¶¶ 3–13, pp. 122–71 (Dr. Kaliner’s statement of qualifications and curriculum vitae); Ex. 2014 ¶¶ 4–11, pp. 85–88 (Dr. Wyse’s statement of qualifications and curriculum vitae).⁸

Patent Owner argues that we should accord “little weight” to Dr. Dyar’s testimony because he has only limited experience working with epinephrine, alkylsaccharides, and intranasal formulations. PO Resp. 10–11. We are not persuaded that this is a reason to discount Dr. Dyar’s testimony. The level of ordinary skill we adopt herein does not require this specific

⁸ Patent Owner asserts that Dr. Smyth’s Reply Declaration (Ex. 1083) “hardly differs from Dr. Smyth’s 290-page first declaration, except where it parrots Dr. Dyar’s declaration.” Sur-reply 10 n.2. We observe that both parties use multiple declarants, and in many cases, duplicative citations (e.g., citing to two of their own declarants, who provide essentially identical opinions). Petitioner also at times cites to an excessive number of declaration paragraphs at one time. See PO Resp. 39 n.5. “Although parties are given wide latitude in how they present their cases,” the parties are reminded that in the Board’s experience, parties should “focus on concise, well-organized, easy-to-follow arguments supported by readily identifiable evidence of record.” PTAB Consolidated Trial Practice Guide (Nov. 2019), at 39; *available at* <https://www.uspto.gov/about-us/news-updates/consolidated-trial-practice-guide-november-2019>.

experience. Rather, the level of skill requires, e.g., an advanced degree in pharmaceutical sciences and at least two years of experience with the development and science of intranasal *or* mucosal delivery systems.

Dr. Dyar holds a Ph.D. in Pharmaceutical Science and has over 30 years of experience working, consulting, and teaching in the field of pharmaceutical sciences, including with respect to ophthalmic ointments and sublingual tablets, which are mucosal delivery systems. Ex. 1082 ¶¶ 2, 3, 5–7. We give Dr. Dyar’s testimony appropriate weight in view of his qualifications and experience.

B. Claim Construction

We interpret a claim “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b). Under this standard, we construe the claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” *Id.*

Petitioner raises two terms for construction: “[a] method of treatment” (a portion of claim 1’s preamble), and “provides a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis” (claim 3). Pet. 13–20. Patent Owner address the entire preamble of claim 1; the term Petitioner addresses from claim 3; and two additional terms recited in claim 1 (“administering intranasally to a mammal” and “aqueous”). PO Resp. 16–17.

We address the preamble (claim 1) and “provides a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis” (claim 3), but determine that on this record we need not

address any other claim term. *See Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999) (stating that only claim terms in controversy need be construed and only to the extent necessary to resolve the controversy).

1. “*A method of treatment*” (Claim 1)

Petitioner contends that the portion of claim 1’s preamble reciting “[a] method of treatment” should be construed as non-limiting, because it merely states the purpose or intended use of the formulations recited in the claim, but offers no additional step, limitation, or antecedent basis for the claim. Pet. 13–14. In the event we find this preamble phrase limiting, Petitioner alternatively proposes a construction. *Id.* at 14.

Patent Owner asserts that “[a] method of treatment” should not be construed in isolation, but rather the entire preamble (“[a] method of treatment of a type-1 hypersensitivity reaction in a mammal”) should be construed to mean:

administration of a dose of epinephrine to a mammal that elicits a biological or medicinal response that (i) is being sought by a researcher or healthcare provider; and (ii) is related to preventing the progression or appearance of, or eliminating, one or more symptoms of type-1 hypersensitivity.

PO Resp. 10.

In our Institution Decision, we determined that “[a] method of treatment” is not limiting, and more specifically, does not invoke an efficacy requirement. Dec. 12–14. For example, we found that this phrase does not recite an essential structure or step that “result[s] in a manipulative difference in the steps of the claim.” *Id.* at 13 (quoting *Bristol-Myers Squibb Co. v. Ben Venue Labs., Inc.*, 246 F.3d 1368, 1376 (Fed. Cir. 2001)).

Rather, we found that the body of claim 1 recites (in general terms) intranasal administration of a particular nasal spray in particular amounts to a subset of mammals, and this method is “performed in the same way regardless [of] whether or not the patient experiences” an efficacious result. *Id.* (quoting *Bristol-Myers Squibb Co.*, 246 F.3d at 1375).

In arguing that the entire preamble is limiting, Patent Owner asserts that the portion of the preamble that reads “. . . of a type-1 hypersensitivity reaction in a mammal” provides important context, e.g., it specifies what the claimed method does, “namely, treatment of a type-1 hypersensitivity reaction in a mammal.” PO Resp. 12. Patent Owner also argues that because “the body of the claim directs that the method be performed on someone *in need*,” this “gives life and meaning to the preamble’s statement of purpose.” *Id.* at 12–13. Finally, Patent Owner argues that the preamble is limiting because it provides antecedent basis for the later recitation of the phrase “the mammal.” *Id.* at 13.

Consistent with our preliminary finding in the Institution Decision, on the full trial record, we determine that the preamble, when considered as a whole, has at least some limiting effect. Dec. 13 n.8. In particular, we find that the language “[a] method of treatment of a type-1 hypersensitivity reaction in a mammal . . . in need thereof” defines the intended patient population, i.e., mammals in need of treatment of a type-1 hypersensitivity reaction. *Id.* Thus, contrary to Patent Owner’s contentions, we are not ignoring relevant context provided by the preamble.

This finding does not, however, convert the phrase “[a] method of treatment” into an efficacy requirement. *See, e.g., TomTom, Inc. v. Adolph*, 790 F.3d 1315, 1323 (Fed. Cir. 2015) (finding that although a portion of the

preamble provides necessary structure, this “does not necessarily convert the entire preamble into a limitation, particularly one that only states the intended use of the invention”). We see no basis for Patent Owner’s argument that the preamble requires “administration of a dose of epinephrine to a mammal that elicits a biological or medicinal response.” PO Resp. 10.

Patent Owner draws on *Eli Lilly & Co. v. Teva Pharmaceuticals International GmbH*, 8 F.4th 1331, 1343 (Fed. Cir. 2021), but the claims there have important differences. PO Resp. 12; Sur-reply 22. The preambles in *Eli Lilly* are directed to “[a] method for treating headache,” and the claim bodies recite a step of “administering to the individual an *effective amount*” of the claimed antibody. *Eli Lilly & Co.*, 8 F.4th at 1335 (emphasis added). The Federal Circuit found the preambles limiting because they “provide the only metric by which one practicing the claim could determine whether the amount administered is an ‘effective amount.’” *Id.* Here, in contrast, challenged claim 1 does not recite an “effective amount” limitation, or any other limitation that demands a certain efficacy result. Rather, as stated above, the claim recites administration of nasal spray in particular amounts, such that the method is “performed in the same way regardless [of] whether or not the patient experiences” an efficacious result. *Bristol-Myers Squibb Co.*, 246 F.3d at 1375. Thus, Patent Owner’s argument is not supported by the language of claim 1 itself.

Patent Owner does not persuade us that the preamble of claim 1 incorporates an efficacy requirement, or that it should be construed in accordance with Patent Owner’s proposal. As discussed above, although the entire preamble helps define the intended patient population, i.e., mammals in need of treatment of a type-1 hypersensitivity reaction, we find that

neither the term “A method of treatment,” nor any other language in the preamble or claim, requires a “biological or medicinal response” as recited in Patent Owner’s proposed construction. Rather, we agree with Petitioner that “[a] method of treatment,” in isolation, is non-limiting.

2. “*provides a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis*” (Dependent Claim 3)

Claim 3 recites:

The method of claim 1, wherein:

intranasal administration of the single dose of the aqueous nasal spray pharmaceutical formulation to the mammal ***provides a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis.***

Ex. 1001, 72:33–37 (emphasis added). Petitioner argues that the bolded phrase (“***provides a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis***”) is not limiting because it is a “[c]oncluding statement[] of intended result[]” that “do[es] not result in ‘a manipulative difference in the steps of the claim.’” Pet. 17 (quoting *In re Copaxone Consol. Cases*, 906 F.3d 1013, 1023 (Fed. Cir. 2018) (quoting *Bristol-Myers Squibb Co.*, 246 F.3d at 1376)).

In our Institution Decision, we rejected Petitioner’s argument that claim 3 results in no manipulative difference in the claim steps. Dec. 14. We noted, for example, that claim 3 encompasses administration of a range of epinephrine dosages and formulations, and that Petitioner had not established that all of the claimed doses and formulations would provide the claimed “plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis” in all claimed mammals. *Id.* at 14–15.

Accordingly, we found the phrase “provides a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis” limiting. *Id.* at 16. We then proceeded to provide a preliminary construction for the term “efficacious” within this phrase. *Id.*

Post-institution, the parties provide proposed constructions for “efficacious,” which differ somewhat from the Board’s preliminary construction. *See* PO Resp. 18–19; Reply 5–6. But no matter what “efficacious” means, both parties acknowledge that at a minimum, claim 3 requires “a plasma epinephrine concentration.” *See, e.g.,* PO Resp. 18 (Patent Owner’s proposed construction begins with “elicits a plasma epinephrine concentration”); Reply 5–6 (discussing “the plasma concentration required to meet claim 3”). Indeed, claim 3 plainly recites that the method “*provides a plasma epinephrine concentration* that is efficacious for the treatment of anaphylaxis.” Ex. 1001, 72:33–37 (emphasis added). On the full trial record, we determine that to resolve the issues presented herein, we need not construe the meaning of “efficacious” in claim 3.

C. Overview of Asserted Prior Art

1. Potta (Ex. 1004)

Potta, titled “Epinephrine Spray Formulations,” is “directed to methods of treating anaphylaxis by administering epinephrine spray formulations to subjects in need of such treatments.” Ex. 1004, code (54), ¶ 1. Potta explains that “[w]hile there are various epinephrine formulations currently available, there remains a need in the art for a quick-onset spray formulation.” *Id.* ¶ 6. Potta discloses that administration of its sprays “preferably . . . occurs via the intranasal route or sublingual route.” *Id.* ¶ 56.

Potta claim 21 relates to aqueous intranasal formulations that include about 0.1% w/w to about 15% w/w epinephrine, and Table 1 exemplifies aqueous formulations comprising 1% w/w epinephrine. *Id.* at claims 1, 19, 21, Table 1. Potta also discloses unit-dose devices that deliver about 100 μ L or 200 μ L of a formulation in a single spray. *Id.* ¶¶ 236–37, 242.

Potta details a clinical study wherein a 6 mg intranasal epinephrine formulation was administered to minipigs. *Id.* ¶¶ 258–60. This study compared the pharmacokinetics of the 6 mg epinephrine intranasal formulation (designated “#M1”) and a 0.3 mg intramuscular formulation. *Id.* ¶ 260, Tables 56, 57. Potta reports that “[w]hen epinephrine was formulated in #M1 and administered intranasally, a more rapid absorption of epinephrine was observed” compared to the intramuscular administration. *Id.* ¶ 260; *see also id.* at Table 57 (reporting plasma concentrations over time for intramuscular and intranasal administration).

Potta also details another clinical study wherein 3 mg and 6 mg intranasal epinephrine formulations were administered to humans. *Id.* ¶¶ 265–68. This study assessed the bioavailability of 3 mg and 6 mg epinephrine intranasal aqueous and hydroalcoholic formulations compared to a 0.3 mg intramuscular formulation (EpiPen). *Id.* ¶ 266. Potta reports pharmacokinetic profiles for the 6 mg and intramuscular formulations both pre- and post-exposure to seasonal allergens, and concludes that “subjects administered 6 milligrams of epinephrine in a hydroalcoholic formulation of the present invention in the presence of an allergen had a higher C_{max} than all other cohorts.” *Id.* ¶ 266; *see also id.* ¶¶ 267–68, Tables 63–65, Figs. 1–3. Potta concludes that “the bioavailability of intranasal formulations of the present invention is as good or better than that of the intramuscular injection

both in subject[s] that are and are not experiencing seasonal allergies.” *Id.* ¶ 269.

2. *Maggio (Ex. 1005)*

Maggio, titled “Compositions for Drug Administration,” relates “to compositions containing a pharmaceutically active ingredient and an alkylsaccharide.” Ex. 1005, code (54), ¶ 3. Maggio explains that although therapeutic agents are often combined with surfactants, surfactants can irritate mucosal membranes and denature proteins. *Id.* ¶ 5. Maggio describes an ideal surfactant as one that stabilizes a therapeutic agent, is non-toxic and non-irritable, has antibacterial activity, and enhances the absorption of the therapeutic agent through membrane barriers. *Id.*

Maggio describes compositions that include alkyl glycosides and/or saccharide alkyl esters and a therapeutic agent, and methods of administering the compositions, including via nasal delivery. *Id.* ¶ 12. Maggio discloses DDM as a preferred alkyl glycoside. *Id.* ¶ 48. Maggio provides clinical data including from a pharmacokinetic comparison of sumatriptan intranasal formulations with and without DDM, and from the administration of phenylephrine oral tablets with DDM in canines. Ex. 1005 ¶¶ 353–56, 358–69, Fig. 12.

3. *Srisawat (Ex. 1006)*

Srisawat is titled “A preliminary study of intranasal epinephrine administration as a potential route for anaphylaxis treatment.” Ex. 1006, 1. Srisawat states that “[t]he intranasal (IN) administration of epinephrine could be an alternative route for anaphylaxis treatment,” but that intranasal epinephrine absorption data in humans are lacking. *Id.* Srisawat thus describes a study wherein various epinephrine doses (0.3, 0.6, 1.25, 2.5, and

5 mg) were administered intranasally to five healthy human adults. *Id.* at 2. The intranasal epinephrine sprays were freshly prepared by dissolving epinephrine bitartrate in saline. *Id.* In a separate arm of the study, the subjects were administered an intramuscular (IM) injection of 0.3 mg epinephrine. *Id.* Plasma epinephrine levels at baseline and up to 120 minutes after administration were determined. *Id.*

We reproduce Srisawat Figure 1 below:

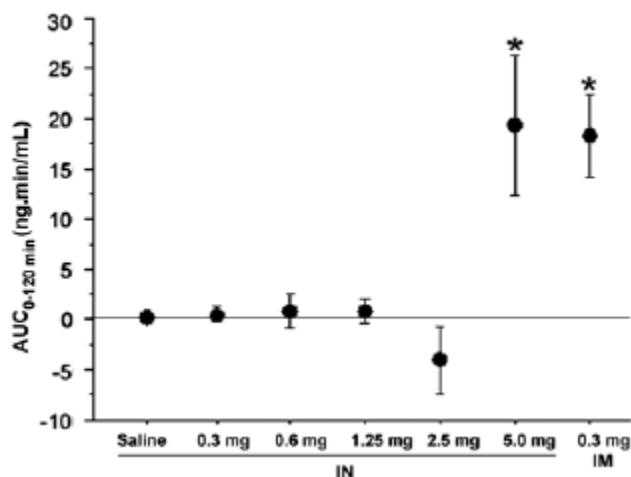


Figure 1. The area under the plasma concentration versus time curve at time 0 – 120 minutes ($AUC_{0-120 \text{ min}}$) of epinephrine absorption at different dosages and routes of administration. The plots are shown as mean $\pm$ SEM. An asterisk indicates a significant difference compared with the IN saline ($p < 0.05$, Mann-Whitney test).

Figure 1 compares the AUC values for each of the epinephrine formulations administered in the study. Consistent with the data reported in Figure 1, Srisawat states that “[s]ignificant systemic absorption of epinephrine via IN

route was observed only at the dose of 5 mg, and the absorption thereof was comparable to that of IM epinephrine.” *Id.* at 1.

Srisawat observed that the plasma epinephrine levels of the IM group “seemed to increase early (as early as 5–10 minutes),” while those of the IN group “seemed to elevate more slowly (about 15–20 minutes after administration).” *Id.* at 4. Srisawat explains that the slower epinephrine absorption through the intranasal route “is not unexpected, as the drug has to diffuse across the nasal epithelial barrier to reach the systemic circulation.” *Id.* According to Srisawat, this could “explain a much higher IN dose (5 mg) required to achieve systemic absorption comparable to the IM route at 0.3 mg.” *Id.*

Srisawat recognizes that the formulation could affect intranasal absorption, and suggests that “[u]sing bile salt or surfactant as a vehicle might also promote the nasal absorption of epinephrine because they have been shown to work well in a dog model.” *Id.* at 4, 5.

4. *Munjal (Ex. 1020)*

Munjal is titled “A Randomized Trial Comparing the Pharmacokinetics, Safety, and Tolerability of DFN-02, an Intranasal Sumatriptan Spray Containing a Permeation Enhancer, With Intranasal and Subcutaneous Sumatriptan in Healthy Adults.” Ex. 1020, 1. Munjal discloses clinical studies that evaluate “the impact of 1-O-n-Dodecyl- β -D-Maltopyranoside (DDM, Intravail A-3TM), a permeation enhancer,” on the absorption profile for intranasally-administered sumatriptan. In particular, Munjal compares the pharmacokinetics of DFN-02 (“a novel intranasal agent comprised of sumatriptan 10 mg plus 0.20% DDM”), with that of commercially available intranasal sumatriptan 20 mg. *Id.* at 1, 3. Munjal

states that absorption “was markedly more rapid” with DFN-02 than with the sumatriptan 20 mg product. *Id.* at 5, Fig. 1, Table 1.

D. Alleged Anticipation

Petitioner asserts that Potta anticipates claims 1–18 under 35 U.S.C. § 102. Pet. 36. Patent Owner opposes. *See* PO Resp. 20–37.

For the reasons discussed below, we determine that Petitioner has shown by a preponderance of the evidence that Potta anticipates claims 1, 2, and 7–17, but not claims 3–6 and 18.

1. Legal Standards

“To anticipate, [a] reference must not only disclose all elements of the claim within the four corners of the document, but must also disclose those elements arranged as in the claim.” *Sanofi-Synthelabo v. Apotex, Inc.*, 550 F.3d 1075, 1083 (Fed. Cir. 2008) (internal quotation omitted). However, “a reference need not always include an express discussion of the actual combination to anticipate.” *Blue Calypso, LLC v. Groupon, Inc.*, 815 F.3d 1331, 1344 (Fed. Cir. 2016). “Instead, a reference may still anticipate if that reference teaches that the disclosed components or functionalities may be combined and one of skill in the art would be able to implement the combination.” *Id.* “[T]he dispositive question regarding anticipation [i]s whether *one skilled in the art* would reasonably understand or infer from the [prior art reference’s] teaching’ that every claim element was disclosed in that single reference.” *Dayco Prods., Inc. v. Total Containment, Inc.*, 329 F.3d 1358, 1368 (Fed. Cir. 2003) (quoting *In re Baxter Travenol Labs.*, 952 F.2d 388, 390 (Fed. Cir. 1991)) (emphasis and alterations in original).

A reference must disclose every limitation of the claimed invention, either expressly or inherently. *Blue Calypso, LLC*, 815 F.3d at 1341. “To

establish that a prior art reference inherently—rather than expressly—discloses a claim limitation, ‘the limitation at issue necessarily must be present, or [is] the natural result of the combination of elements explicitly disclosed by the prior art.’” *Endo Pharms. Sols., Inc. v. Custopharm Inc.*, 894 F.3d 1374, 1381 (Fed. Cir. 2018) (alteration in original) (quoting *Par Pharm., Inc. v. TWI Pharms., Inc.*, 773 F.3d 1186, 1196 (Fed. Cir. 2014)). “Inherency may not be established by probabilities or possibilities. The mere fact that a certain thing *may* result from a given set of circumstances is not sufficient.” *Continental Can Co. USA, Inc. v. Monsanto Co.*, 948 F.2d 1264, 1269 (Fed. Cir. 1991) (citations omitted). “If, however, the disclosure is sufficient to show that the natural result flowing from the operation as taught would result in the performance of the questioned function, it seems to be well settled that the disclosure should be regarded as sufficient.” *Id.*

2. Analysis

a) Claim 1

We find that Petitioner demonstrates by a preponderance of the evidence that Potta anticipates claim 1. Pet. 30–34. Petitioner demonstrates that Potta teaches administering aqueous nasal spray pharmaceutical formulations to humans in need of treatment of type-1 allergic reactions, including anaphylaxis. *See, e.g.*, Pet. 30–32; Ex. 1004 (Potta) ¶¶ 1, 56, 69, 242, Table 1, claims 1, 19, 21.

Petitioner also demonstrates that Potta teaches claim 1’s recitations of “about 0.1 mg to 2.4 mg of epinephrine” in a “single dose.” Ex. 1001, 72:21–23. In particular, Potta teaches 1% w/w aqueous epinephrine formulations in Table 1, and a preference for intranasal administration.

Ex. 1004 (Potta), Table 1, ¶ 56. Potta's claim 21 echoes these disclosures by reciting intranasal administration of aqueous epinephrine formulations comprising "from about 0.1% w/w/ to about 15% w/w epinephrine." *Id.* at claims 1, 19, 21. Potta also teaches unit-dose devices for intranasal administration of epinephrine that deliver a single dose of about 100 µL or 200 µL in a single spray. *Id.* ¶¶ 56–58, 236–37; Pet. 32–34. Based on these disclosures, Petitioner establishes that a person of ordinary skill in the art would have understood Potta as disclosing intranasal epinephrine formulations in Table 1 that correspond to 1 mg/100 µL or 2 mg/200 µL in a single dose, and intranasal epinephrine formulations in claim 21 that correspond to about 0.1–15 mg in a single 100 µL spray or 0.2–30 mg in a single 200 µL spray. Pet. 33–34; Ex. 1002 (Smyth Decl.) ¶¶ 240–51, 261; Ex. 1003 (Weinstein Decl.) ¶¶ 263–65, 275. These epinephrine doses fall within or overlap with those recited in challenged claim 1.

Patent Owner argues that Potta does not anticipate because it does not disclose actual administration of the claimed doses, but instead discloses only administration of higher doses. PO Resp. 21 (citing Ex. 2013 (Kaliner Decl.) ¶ 146; Ex. 2014 (Wyse Decl.) ¶ 128), 24–26. This argument is unavailing because "anticipation does not require actual performance of suggestions in a disclosure;" "[r]ather, anticipation only requires that those suggestions be enabling to one skilled in the art." *Bristol-Myers Squibb Co.*, 246 F.3d at 1379 (affirming anticipation where the anticipatory reference suggested the claimed step of premedicating patients, but did not actually premedicate patients); *see also* Reply 7–8. Potta teaches methods of treating anaphylaxis by administering the disclosed formulations to patients, and discloses administering 3 mg and 6 mg intranasal formulations to adults with

seasonal allergies. *See, e.g.*, Ex. 1004 (Potta) ¶¶ 1, 56, 265; Ex. 1002 (Smyth Decl.) ¶¶ 241, 244, 246–48, 256–61; Ex. 1083 (Smyth Reply Decl.) ¶¶ 44–45, 51; Ex. 1003 (Weinstein Decl.) ¶¶ 142–44, 148–49. This is sufficient to anticipate.⁹ *Bristol-Myers Squibb Co.*, 246 F.3d at 1378–79.

Patent Owner asserts that “[a]t best, Potta discloses testing for the stability and spray plume characteristics of the Table 1 formulations.” PO Resp. 23 (citing Ex. 1004 (Potta) ¶¶ 243–51; Ex. 2013 (Kaliner Decl.) ¶ 146; Ex. 2014 (Wyse Decl.) ¶ 133). This argument is unavailing, because it reads Potta too narrowly. Merely because Potta discloses that the Table 1 formulations were subjected to certain testing doesn’t mean that a person of ordinary skill in the art would have understood Potta as indicating that such formulations are useful solely for such testing. For example, after reporting the stability results, Potta concludes that the “stability characteristics of the formulations of the present invention will allow the formulations to be effective when used by patients.” Ex. 1004 (Potta) ¶ 244. In other words,

⁹ In a footnote, Patent Owner asserts that “Potta Claim 21 is not enabled for dose amounts below 3 mg.” PO Resp. 25 n.4. Patent Owner fails to adequately develop this argument, but to the extent Patent Owner is arguing that Potta claim 21 is not enabled because Potta did not actually administer doses below 3 mg, actual administration is not required for enablement. *See Bristol-Myers Squibb Co.*, 246 F.3d at 1379–80 (finding that prior art’s “pretreatment suggestion was enabling,” even though the reference did not disclose actual premedication of patients); Reply 9. Moreover, Potta is presumptively enabled, and Patent Owner has not rebutted this presumption. *See Amgen, Inc. v. Hoechst Marion Roussel, Inc.*, 314 F.3d 1313, 1355 (Fed. Cir. 2003) (noting a “presumption . . . that both the claimed and unclaimed disclosures in a prior art patent are enabled”); *In re Antor Media Corp.*, 689 F.3d 1282, 1288 (Fed. Cir. 2012) (extending presumption to prior art printed publications).

Potta teaches that the testing was performed to evaluate the suitability of the Table 1 formulations for administration to patients.

Patent Owner additionally asserts in a footnote that an earlier application in Potta's priority chain, "which is entitled 'Sublingual Epinephrine Spray,' makes clear that the formulations in Table 1 were intended for sublingual use, not intranasal use."¹⁰ PO Resp. 24 n.3 (citing, e.g., Ex. 2013 (Kaliner Decl.) ¶ 148; Ex. 2014 (Wyse Decl.) ¶ 133; Ex. 2006 (Potta priority app.) ¶ 71). Potta, a continuation-in-part of the priority application Patent Owner cites, adds disclosure regarding intranasal administration, which is missing from the priority application. *Compare*, e.g., Ex. 1004 (Potta) ¶ 56 ("preferably administration occurs via the intranasal route or sublingual route"), *with* Ex. 2006 (Potta priority app.) ¶ 16 (disclosing sublingual administration of epinephrine spray). We agree with Dr. Smyth that this additional disclosure "confirms that the [Potta] inventors intended that the formulations of Table 1 be administered via sublingual *and* intranasal epinephrine administration." Ex. 1083 (Smyth Reply Decl.) ¶ 46 (emphasis added); *see also id.* ¶¶ 47–48; Reply 8.

Patent Owner also argues that Potta does not anticipate claim 1 because "it is clear that Potta does not disclose *any* 'biological or medicinal response' that a researcher or healthcare provider would have been seeking related to type-1 hypersensitivity based on administration of an intranasal dose of epinephrine of less than 3 mg." PO Resp. 27 (citing Ex. 2013

¹⁰ Patent Owner incorrectly characterizes this earlier application as a provisional application. *See* PO Resp. 24 n.3. Exhibit 2006 is a publication of one of the non-provisional applications in Potta's priority chain. *See* Ex. 2006, codes (10), (21).

(Kaliner Decl.) ¶ 154; Ex. 2014 (Wyse Decl.) ¶ 137). This argument is unavailing because it is based on Patent Owner’s proposal to import an efficacy requirement into claim 1’s preamble, which we reject above. *See supra* Section II.B.1.

On Sur-reply, Patent Owner newly argues that a person of ordinary skill in the art reading Potta’s Table 1 and claim 21 would “at best, ‘at once envisage’ only the use of a 3 mg dose or a 6 mg dose within the broad set of formulations disclosed in Potta Table 1 and Potta Claim 1,” and “would not ‘at once envisage’ epinephrine doses that were never administered intranasally or shown to have any biological effect or result.” Sur-reply 5–6, 20. We decline to consider this argument, because it was not timely presented in the Patent Owner Response. *See* Dec. 45 (“The Board will deem waived any issue not raised in a timely response to the Petition.”); Paper 13 (Scheduling Order), 9 (“Patent Owner is cautioned that any arguments not raised in the response may be deemed waived.”).

However, even if we were to consider Patent Owner’s untimely argument, it is unavailing. First, Patent Owner’s invocation of the “at once envisage” line of cases is misdirection because Petitioner did not rely on this legal theory when arguing that Potta discloses the claimed dosages. *See* Pet. 32–34. Second, the Sur-reply does not support this argument with citation to any support from the perspective of a person of ordinary skill in the art. *See Johnston v. IVAC Corp.*, 885 F.2d 1574, 1581 (Fed. Cir. 1989) (noting that attorney argument is “no substitute for evidence”). Third, Patent Owner narrowly focuses on Potta’s Table 1 and claim 21, while ignoring additional relevant disclosures such as Potta’s teachings regarding unit-dose devices that deliver 100 µL and 200 µL single doses, as discussed above. Finally,

Patent Owner cites *Mylan Pharmaceuticals Inc. v. Merck Sharp & Dohme Corp.*, 50 F.4th 147 (Fed. Cir. 2022). In that case, the court found no error in the Board’s finding that a class of 957 compounds “is insufficient to meet the ‘at one envisage’ standard.” *Id.* at 154. Patent Owner fails to explain how those facts parallel the situation here, where Petitioner asserts, for example, that the 1% w/w epinephrine formulations in Potta’s Table 1, delivered via Potta’s 100 μ L single-dose spray devices, results in a discrete dosage of 1 mg epinephrine, which falls within claim 1. Pet. 33.

For the above reasons, on the full trial record we determine that Petitioner has shown by a preponderance of the evidence that Potta anticipates claim 1.

b) Dependent Claims 2 and 7–17

Dependent claims 2 and 7–17 depend directly or indirectly from claim 1. Petitioner argues that Potta discloses the additional limitations introduced in each of these dependent claims, and supports its argument with citations to specific portions of Potta and testimony from its declarants. *See* Pet. 34, 36–39, 40–42. Other than its arguments over the subject matter of claim 1 incorporated into these dependent claims, which we discussed above, Patent Owner proffers no arguments that Potta fails to describe the limitations of these dependent claims. *See generally* PO Resp. Accordingly, Patent Owner has forfeited any anticipation arguments related to these claims. *See In re NuVasive, Inc.*, 841 F.3d 966, 974–75 (Fed. Cir. 2016); Paper 13, 9 (“Patent Owner is cautioned that any arguments not raised in the response may be deemed waived.”).

Based on our review of Petitioner’s arguments and evidence relating to dependent claims 2, 7–10, and 12–17 (*see* Pet. 34, 36–39, 40–42), which

we adopt as our own, and our analysis of claim 1 in Section II.D.2.a above, we find that Petitioner has proved by a preponderance of the evidence that Potta anticipates these claims.

Regarding claim 11, in the Institution Decision we expressed reservations regarding the sufficiency of Petitioner’s arguments on this claim. Dec. 26–28. We observed that “Petitioner does not assert that Potta expressly discloses using BKC as an absorption enhancer,” and expressed reservations regarding Petitioner’s argument that a person of ordinary skill in the art would have “immediately envisioned” using BKC as an absorption enhancer in Potta’s formulations. *See id.* Post-institution, both parties clarify that Potta does in fact disclose BKC as an absorption enhancer. *See* Ex. 1004 (Potta) ¶ 20; Ex. 1083 (Smyth Reply Decl.) ¶ 88; Ex. 2014 (Wyse Decl.) ¶ 110; Ex. 1080 (Kaliner depo.), 174:18–175:22. Potta also discloses using amounts of permeation enhancers¹¹ that encompass the claimed range. Ex. 1004 (Potta) ¶ 85. Patent Owner does not specifically dispute Petitioner’s anticipation arguments concerning claim 11. In view of Potta’s disclosures and the arguments of record, we conclude that Petitioner has proved by a preponderance of the evidence that Potta anticipates claim 11. Pet. 40; Reply 16; Ex. 1083 ¶¶ 84–90.

c) Dependent Claims 3–6 and 18

Claim 3 depends from claim 1 and recites that the intranasal administration of the single dose of the claimed formulation “provides a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis.” Ex. 1001, 72:33–37. Thus, the plain language of claim 3

¹¹ We use the terms “permeation enhancer” and “absorption enhancer” interchangeably. *See* Ex. 1001, 34:57–64.

requires, at a minimum, that intranasal administration “provides a plasma epinephrine concentration.”

There is no dispute that Potta does not disclose actual administration of the formulations Petitioner relies on for anticipation, and thus does not disclose any information on whether their administration provides a plasma epinephrine concentration. *See, e.g.*, PO Resp. 6; Ex. 2041 (Dyar depo.), 66:20–68:2. Indeed, at deposition, Petitioner’s declarants were unable to articulate what the plasma epinephrine concentration of Potta Table 1 and Claim 21 formulations would be if they were administered. *See* Ex. 2018 (Smyth depo.), 153:4–13; Ex. 2017 (Weinstein depo.), 127:14–128:2, 129:11–19.

Instead, Petitioner relies on a theory of inherent anticipation to meet this limitation of claim 3, arguing that a person of ordinary skill in the art “would understand that a plasma concentration that is efficacious for the treatment of anaphylaxis would occur upon administration of the Potta formulations,” and thus the limitation of claim 3 “is an inherent property of practicing the claimed method.” *See* Pet. 35 (citing Ex. 1002 (Smyth Decl.) ¶¶ 270–73; Ex. 1003 (Weinstein Decl.) ¶¶ 286–87; *Perricone v. Medicis Pharm. Corp.*, 432 F.3d 1368, 1377–78 (Fed. Cir. 2005)). We find that the Petition and cited declarant testimony is deficient for failing to demonstrate why Potta’s formulations would *necessarily* result in a plasma epinephrine concentration. *See id.*; *see also* Ex. 1002 (Smyth Decl.) ¶¶ 270–73; Ex. 1003 (Weinstein Decl.) ¶¶ 286–87).

Dr. Smyth merely asserts that “the functional effect of ‘provid[ing] a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis’ is an inherent property of the 1% w/w epinephrine formulations

in Table 1 and the formulations of Potta claim 21 (i.e., ‘from about 0.1% w/w to about 15% w/w epinephrine or a salt thereof’), because such formulations would elicit the desired biological or medicinal response.” Ex. 1002 ¶ 272. Dr. Weinstein provides an essentially identical opinion. Ex. 1003 ¶ 286. Neither declarant supports their opinion with any explanation as to why the Potta formulations would provide the claimed plasma epinephrine concentration or elicit a biological or medicinal response. Accordingly, we do not credit these opinions, and determine that Petitioner has not met its burden to establish that Potta inherently anticipates claim 3. *See n re Am. Acad. of Sci. Tech Ctr.*, 367 F.3d 1359, 1368 (Fed. Cir. 2004) (“[T]he Board is entitled to weigh the declarations and conclude that the lack of factual corroboration warrants discounting the opinions expressed in the declarations.”).

In its Reply brief, Petitioner argues that “[u]nder inherent anticipation, Petitioner is not required to know the specific plasma epinephrine concentration upon administration; it is the natural consequence of administration.” Reply 14. We agree that in this case Petitioner need not identify the specific plasma epinephrine concentration that would result from administration of Potta’s formulations. Petitioner, however, has not demonstrated that the administration of Potta’s formulations would *necessarily* result in any plasma epinephrine concentration at all. Indeed, the record demonstrates that not all intranasal epinephrine formulations result in a plasma epinephrine concentration. For example, Srisawat describes a study wherein various epinephrine doses (0.3, 0.6, 1.25, 2.5, and 5 mg) were administered intranasally to five healthy human adults, but only the 5 mg formulation showed “[s]ignificant systemic absorption of epinephrine”

compared to the saline control. Ex. 1006 (Srisawat), 1, 2, Figure 1. Even the '414 patent reports that “Srisawat et al. observed no blood level of epinephrine at the intranasal dose level of 2.5 mg and below.” Ex. 1001, 15:18–20.

On Reply, Petitioner tries to salvage the conclusory inherency argument offered in the Petition by newly arguing that Junior and Srisawat demonstrate that administration of Potta’s formulations will necessarily result in a plasma epinephrine concentration. Reply 15. Petitioner supports its arguments in part by citation to an entirely new declarant introduced on Reply. *See id.* (citing, e.g., Ex. 1082 (Dyar Reply Decl.)).

Petitioner’s belated reliance on Junior and Srisawat is improper.¹² An IPR petition must identify “with particularity . . . the evidence that supports the grounds for the challenge to each claim.” 35 U.S.C. § 312(a)(3). As explained in our Trial Practice Guide, “Petitioner may not submit new evidence or argument in reply that it could have presented earlier, e.g. to make out a prima facie case of unpatentability.” Trial Practice Guide, 73. Here, Petitioner could have discussed Junior and Srisawat in the Petition to support its inherency argument, but did not.¹³ *See* Pet. 35. Petitioner’s new

¹² Patent Owner argues that Petitioner’s reliance on references outside of Potta to demonstrate inherent anticipation converts Petitioner’s anticipation challenge to an obviousness challenge. PO Resp. 29, 36. This argument is contrary to law. *See, e.g., Continental Can Co.*, 948 F.2d at 1268 (“To serve as an anticipation when the reference is silent about the asserted inherent characteristic, such gap in the reference may be filled with recourse to extrinsic evidence.”).

¹³ To the extent Petitioner argues that its reliance on Junior and Srisawat is driven by analysis in the Institution Decision or by Patent Owner’s arguments in the Patent Owner Response, we disagree. Petitioner has

post-institution inherency arguments based on Junior and Srisawat—both of which were submitted with the Petition—come too late.

However, even if we were to consider Petitioner’s inherency arguments based on Junior and Srisawat, we nevertheless find these references insufficient to demonstrate inherent anticipation. Petitioner essentially argues that Junior and Srisawat indicate that a plasma epinephrine concentration results from low-dose intranasal epinephrine, and thus the Potta formulations containing absorption enhancers would “[c]ertainly” provide a plasma epinephrine concentration. Reply 15 (citing Ex. 2017, 129:21–130:4; Ex. 1083 (Smyth Reply Decl.) ¶¶ 72, 78–83; Ex. 1082 (Dyar Reply Decl.) ¶¶ 85–99, 105–06, 108–09; Ex. 2017 (Weinstein depo.), 126:12–22, 129:21–130:4).

As an initial matter, Petitioner’s reliance on the inclusion of any absorption enhancers in the Potta formulations to argue inherency is improper because Petitioner’s anticipation arguments for claim 3 (and base claim 1) in the Petition nowhere discuss the contribution of an absorption enhancer. This is underscored by Petitioner’s citations in the Reply, which cite only *new* declarant testimony submitted on Reply. *See* Reply 15 (citing Ex. 1083 (Smyth Reply Decl.) ¶ 72; Ex. 1082 (Dyar Reply Decl.) ¶ 106).

As to Junior, we agree with Patent Owner that Petitioner’s reliance on this reference “is a bridge too far,” given the very different epinephrine administration described in Junior. Sur-reply 8; Ex. 1080 (Kaliner depo.),

recognized from the start that claim 3 requires “an ‘efficacious’ plasma epinephrine concentration.” Pet. 9. Even if the contours of the term “efficacious” are disputed, Petitioner understood at the Petition stage that claim 3 requires some plasma epinephrine concentration.

206:10–207:6. Junior describes the placement of lidocaine and epinephrine-soaked cotton swabs in direct contact with the nasal cavities for 40 minutes, for the purpose of preventing bleeding during nasal surgery. Sur-reply 8; Ex. 1082 (Dyar Reply Decl.) ¶ 95; Ex. 1083 (Smyth Reply Decl.) ¶¶ 63–65. According to Petitioner, Junior reported that doses of epinephrine lower than the amount in the 1% w/w Potta formulations, when applied topically to the nose, still caused an increase in plasma epinephrine concentration. Ex. 1083 (Smyth Reply Decl.) ¶¶ 97–99 (citing, e.g., Ex. 1066 (Junior), 1, 6, 8); Ex. 1082 (Dyar Reply Decl.) ¶ 66.

Petitioner’s argument is unavailing, because Petitioner fails to adequately address the different method and timing of epinephrine administration in Junior versus that taught in Potta. *See* Sur-reply 8. Junior placed cotton swabs soaked in lidocaine and epinephrine in direct contact with the nostril for up to 40 minutes. *See* Ex. 1066 (Junior), 1–2; Ex. 1082 (Dyar Reply Decl.) ¶ 95. The Potta formulations, in contrast, comprise a single dose nasal spray. As aptly summarized by Patent Owner’s counsel, “you have a completely different mode of delivery of epinephrine, you have different amounts of epinephrine being used over different periods of time, you have those being delivered to a different patient population, and you have this confounding factor of lidocaine.” Tr. 33:18–21; Sur-reply 8. Petitioner’s declarant Dr. Dyar admitted that pharmacokinetic profiles may differ based on a change the route of administration, formulation, or dose of a drug. *See* Ex. 2041 (Dyar depo.), 51:17–53:22; *see also* Reply 18 (“POSAs knew that different delivery systems resulted in varying absorption levels”). And although the amount of epinephrine in Junior is much less than that in Potta, given both the different modes of delivery and different

contact times, we are unable, on this record, to conclude that the differences “wash[] away when you talk about magnitudes of difference in amounts.” Tr. 17:6–10.

We also find that Petitioner’s reliance on Srisawat to demonstrate inherency is unavailing. As discussed above, Srisawat intranasally administered epinephrine in saline at doses of 0.3, 0.6, 1.25, 2.5, and 5 mg, but expressly states that “[s]ignificant systemic absorption of epinephrine via IN [intranasal] route was observed only at the dose of 5 mg.” Ex. 1006 (Srisawat), 1; *see also id.* at Fig. 1; PO Resp. 36. Accordingly, Srisawat expressly teaches that not all intranasal epinephrine formulations result in a plasma epinephrine concentration, thus directly contradicting Petitioner’s suggestion that we can assume that administration of Potta’s formulations would inherently provide a plasma epinephrine concentration.

Nevertheless, Petitioner contends that after administration of epinephrine in a saline-only formulation, Srisawat’s subjects showed “transient tremor,” “palpitation,” and “increases in heart rate, diastolic and systolic blood pressure at $T_{\max}$.” Reply 15. In connection with its arguments on whether Potta anticipates claim 3, Petitioner does not demonstrate that these biological effects necessarily indicate that administration of the epinephrine provided a plasma epinephrine concentration. *See id.* at 14–15. Nevertheless, Petitioner appears to be asking us to assume that these effects must have been caused by epinephrine absorption. On this record, we are unable to do so.

First, relying on these side effects as a proxy for a plasma epinephrine concentration is flatly inconsistent with Srisawat’s teaching that intranasal epinephrine at 0.3, 0.6, 1.25, and 2.5 mg resulted in no significant difference

of systemic absorption of epinephrine as compared to saline. Ex. 1006 (Srisawat), 1, 4 (Fig. 1).¹⁴ Second, we agree with Patent Owner that “Srisawat does not indicate which subjects had these responses, and it is clear they were not observed in all subjects.” PO Resp. 32; Ex. 1006 (Srisawat), 4 (“one” subject experienced transient tremor; “two” subjects experienced palpitation; “most” but not all subjects experienced heart rate and blood pressure increases); Sur-reply 9.

In short, “Srisawat has not provided enough data to conclude that these effects resulted from the administration of intranasal epinephrine at any specific dose, let alone doses less than 5 mg.” Ex. 2013 (Kaliner Decl.) ¶ 157. Petitioner’s declarants conceded this point. *See* Ex. 2017 (Weinstein depo.), 133:2–9 (acknowledging that “Srisawat didn’t say exactly which doses people had adverse effects”); Ex. 2041 (Dyar depo.), 78:22–79:6 (“because we don’t have the actual individual patient responses, . . . I cannot answer which subjects specifically . . . would not have had an increase in heart rate”); *see also* Tr. 19:16–23 (when asked “[H]ow do we know that even the 0.3 milligram dosage resulted in increase in heart rate, increase in blood pressure in most subjects?”, Petitioner’s counsel conceded “I don’t think we know for sure. You know, Srisawat tells us what it tells us and is

¹⁴ Petitioner’s counsel indicates that Srisawat’s Figure 1 is on a nanogram scale whereas other articles are on a picogram scale, making it harder to tell if there was measurable epinephrine absorption. *See* Tr. 18:2–6. Nevertheless, we agree with Patent Owner’s counsel that “[W]hether we’re talking about nanograms or picograms, we can only go with the information that Srisawat gives us, which is to say that there’s no measurable plasma epinephrine concentration within the arrow marks that are shown for the 0.3, 0.6, 1.25, and 2.5 milligram doses in Srisawat.” *Id.* at 32:9–13.

certainly imperfect . . .”). For at least these reasons, we are not persuaded that Srisawat assists in demonstrating that the missing disclosure is necessarily present in Potta. The mere *possibility* that administering the formulations of Potta would produce the claimed plasma epinephrine concentration is not enough. *Continental Can Co.*, 948 F.2d at 1269 (explaining that inherency is not established by “possibilities”).

Although not clear from the face of its Reply, Petitioner’s declarants at times appear to attempt to read out claim 3’s express recitation of “provides a plasma epinephrine concentration” by arguing that the “‘plasma epinephrine concentration’ in the Board’s construction does not require a measurable level of plasma epinephrine concentration above baseline.” Ex. 1082 (Dyar Reply Decl.) ¶ 74; *see also* Ex. 1083 (Smyth Reply Decl.) ¶ 31; Sur-reply 23. We reject this argument. First, it is not clearly explained in Petitioner’s briefs. *See, e.g.*, Reply 15 (stating “*if* the Board’s construction requires a plasma concentration over baseline,” but nowhere clearly taking a position that claim 3 does not require a plasma epinephrine concentration over baseline); *see also* 37 C.F.R. § 42.6(a)(3) (“Arguments must not be incorporated by reference from one document into another document.”). Second, even if we were to consider this argument, we agree with Patent Owner that it “makes little sense and is also unsupported” in the intrinsic record of the ’414 patent. PO Resp. 23. Indeed, the argument improperly reads out claim 3’s express recitation that the “intranasal administration . . . provides a plasma epinephrine concentration.” Ex. 1001, 72:33–37.

For the above reasons, we determine on the full trial record that Petitioner has not demonstrated by a preponderance of the evidence that

Potta anticipates claim 3. Because claims 4–6 and 18 depend directly or indirectly from claim 3, and because Petitioner’s arguments for these claims do not rectify the deficiencies discussed above for claim 3, we similarly determine on the full trial record that Petitioner has not demonstrated by a preponderance of the evidence, that Potta anticipates claims 4–6 and 18.

E. Alleged Obviousness

Petitioner asserts that claims 1–20 are unpatentable under 35 U.S.C. § 103 as obvious over the combination of (1) Potta, Maggio, and Munjal; and (2) Srisawat, Maggio, and Munjal. Pet. 44–60. Patent Owner opposes. PO Resp. 47–60.

For the reasons discussed below, on this record we determine that Petitioner has shown by a preponderance of the evidence that claims 1, 2, 7–9, and 11–17 are unpatentable as obvious over Potta, Maggio, and Munjal, but has not shown that claims 3–6, 10, and 18–20 are unpatentable as obvious over this combination. Additionally, as discussed further below, we limit our consideration of the ground relying on Srisawat, Maggio, and Munjal to claims 3–6 and 18–20, and find that Petitioner has not shown by a preponderance of the evidence that these claims are unpatentable as obvious over Srisawat, Maggio, and Munjal.

1. Legal Standards

A claim is unpatentable as obvious “if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains.” 35 U.S.C. § 103; *see also KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is

resolved based on underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) any objective indicia of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). An obviousness determination requires finding “a motivation to combine accompanied by a reasonable expectation of achieving what is claimed in the patent-at-issue.” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367 (Fed. Cir. 2016).

2. *Alleged Obviousness Over Potta, Maggio, and Munjal*

a) *Claim 1*

We begin by discussing the contentions in the Petition and our findings based on those contentions. At the outset we note that although challenged claim 1 does not expressly recite an absorption enhancer, Petitioner’s obviousness arguments include assertions that a skilled artisan would have been motivated to optimize Potta’s formulations by including DDM as an absorption enhancer. *See, e.g.*, Pet. 49 (“Here, POSAs would have pursued the low dose formulations in Potta and understood them to work with absorption enhancement agents (though none are required by claim 1) to practice the claimed method.”); *id.* at 49–51 (addressing DDM). Thus, although on this record Petitioner has shown by a preponderance of the evidence that Potta teaches each limitation of claim 1 (*see supra* Section II.D.2.a; *see also* Pet. 30–34 (mapping Potta’s disclosures to the limitations of claim 1)), our analysis of Petitioner’s obviousness arguments requires additional considerations.

On this record, we find that given the known drawbacks of intramuscular injections to treat anaphylaxis, skilled artisans would have

been motivated to use alternate means of administration, such as an intranasal epinephrine spray. Pet. 44–45; Ex. 1002 (Smyth Decl.) ¶¶ 86, 366; Ex. 1003 (Weinstein Decl.) ¶¶ 90, 315; *see also* PO Resp. 38 (conceding that skilled artisans were motivated to pursue alternatives to intramuscular epinephrine, including formulations for intranasal delivery). We also find that in developing an intranasal epinephrine spray, skilled artisans would have been motivated to achieve rapid onset of action and an epinephrine dose balanced to account for side effects and efficacy. Pet. 44; Ex. 1002 (Smyth Decl.) ¶¶ 365–68; Ex. 1003 (Weinstein Decl.) ¶¶ 314–25; Ex. 1006 (Srisawat) at 5 (“[D]ifferent epinephrine formulations for nasal spray should also be investigated to discover one . . . which can achieve better and faster systemic absorption.”); Ex. 1010 (Fleming) ¶ 12 (noting that 5 mg epinephrine nasal spray with $T_{\max}$ of 70 ± 17 minutes “is insufficient to be of any utility in anaphylactic shock”); Ex. 1004 (Potta) ¶ 6 (“there remains a need in the art for a quick-onset spray formulation”); Ex. 1080 (Kaliner depo.), 31:13–32:4 (agreeing that rapid onset is very important to treat anaphylaxis).

Petitioner further demonstrates that “[b]ecause treatment of anaphylaxis requires rapid onset to avoid death or complications,” skilled artisans would have been motivated to use an absorption enhancer in an intranasal epinephrine spray. Pet. 6; *see also id.* at 45. Absorption enhancers were known to provide faster onset at lower doses by causing rapid systemic bioavailability. *Id.* at 6; Ex. 1002 (Smyth Decl.) ¶ 99; Ex. 1025 (Cunningham 2018), 1 (“Intravail® excipients have been demonstration to substantially increase systemic absorption (increased $C_{\max}$ and bioavailability) and speed of drug uptake (shortened $T_{\max}$).”); Ex. 1020

(Munjal), 10 (“Adding a permeation enhancer to IN sumatriptan markedly speeds its absorption compared with commercially available IN sumatriptan 20 mg.”). The prior art expressly taught inclusion of absorption enhancers in intranasal epinephrine sprays. *See, e.g.*, Ex. 1004 (Potta) ¶¶ 5 (identifying “a need in the art for a quick-onset spray formulation”), 19–20 (disclosing intranasal formulations with permeation enhancers, including EDTA), Table 1 (formulations include EDTA); Ex. 1006 (Srisawat) 5 (teaching inclusion of surfactant to “promote the nasal absorption of epinephrine”); Ex. 1002 (Smyth Decl.) ¶¶ 88–89, 369–71; Ex. 1003 (Weinstein Decl.) ¶ 322; Ex. 1080 (Kaliner depo.), 174:18–175:22 (discussing permeation enhancers in Potta’s Table 1 formulations).

We also find that given epinephrine’s dose-dependent side effects and narrow therapeutic window, skilled artisans would have been motivated to pursue “low dose epinephrine nasal formulations,” such as the 1% w/w epinephrine (i.e., 1 mg/100 µl) formulations taught in Potta. Pet. 6, 21–22, 49; Ex. 1003 (Weinstein Decl.) ¶¶ 94–107, 316–18; Ex. 1041 (Simons 2006), 6, 8, Fig. 4; Ex. 1014 (Simons 2011), 11; Ex. 1052 (Kemp), 3–4; Ex. 2013 (Kaliner Decl.) ¶ 23) (“Even at recommended doses, . . . epinephrine can present adverse effects.”). Skilled artisans would have known that avoiding side effects would be especially relevant when using absorption enhancers, which as noted above, were well known to increase the uptake and systemic bioavailability of drugs. *See also* Ex. 1002 (Smyth Decl.) ¶¶ 128–30; Ex. 1003 (Weinstein Decl.) ¶ 100; Ex. 2013 (Kaliner Decl.) ¶ 231 (“[M]ore rapid administration of epinephrine . . . may increase the risk of serious adverse effects.”). This motivation to use a low epinephrine dose is also consistent with the general drive of skilled artisans

to provide the lowest effective dose to minimize side effects. Reply 17; Ex. 1080 (Kaliner depo.), 220:6–222:9 (discussing how “we all try to use the lowest effective dose,” and that “lower doses tend to have fewer issues with them;” agreeing that “skilled artisans would generally be motivated to administer the lowest effective dose to minimize side effects”).

Petitioner also demonstrates that a person of ordinary skill in the art would have been motivated with a reasonable expectation of success to select DDM as an absorption enhancer for use in Potta’s formulations. Pet. 45. Although Potta teaches the use of absorption enhancers in its formulations and names a variety of example enhancers (*see, e.g.*, Ex. 1004 ¶¶ 85–86), it does expressly name DDM. The prior art, moreover, taught the combination of DDM and epinephrine to enhance epinephrine absorption across mucosal membranes in ophthalmic, buccal, lingual, and sublingual formulations. Pet. 47–48; Ex. 1002 (Smyth Decl.) ¶¶ 109–10, 375–77; Ex. 1003 (Weinstein Decl.) ¶¶ 352–54; Ex. 1021 (Kee), 9:15–38; Ex. 1022 (Hill), code (57), ¶¶ 53, 102. Maggio also suggested the use of DDM with epinephrine. Pet. 48; Ex. 1002 (Smyth Decl.) ¶ 377; Ex. 1003 (Weinstein Decl.) ¶ 352; Ex. 1005 (Maggio) ¶ 356.

Petitioner demonstrates that DDM was known to be “safe, non-toxic, and non-irritating, while still promoting rapid absorption of small molecules across the nasal mucosa[.]” Pet. 45, 49–51; *see also, e.g.*, Ex. 1002 (Smyth Decl.) ¶¶ 374–75, 377; Ex. 1003 (Weinstein Decl.) ¶¶ 330–38; Ex. 1016 (Maggio 2014), 5–6. Maggio and Munjal also demonstrate the advantages of using DDM (albeit with other drug molecules) to increase uptake and bioavailability across mucosal membranes, allowing for reduced doses and side effects. Pet. 47, 49–51; Ex. 1002 (Smyth Decl.) ¶¶ 378–81, 393–96,

404–06; Ex. 1003 (Weinstein Decl.) ¶¶ 352–53; Ex. 1005 (Maggio) ¶¶ 353–56, 358–69; Ex. 1020 (Munjal), 1, 5–6, 9, 10.

Accordingly, on this record, Petitioner establishes that Potta teaches all limitations of claim 1, and that skilled artisans would have been motivated to optimize the “low dose” intranasal formulations taught in Potta’s Table 1 and claim 21 by adding an absorption enhancer such as DDM, with a reasonable expectation of success. *See, e.g.*, Pet. 44–51.

Before deciding obviousness, we next analyze Patent Owner’s arguments in detail.

(1) Patent Owner’s Arguments Regarding the Claimed Dose Range

Patent Owner makes several arguments questioning whether Petitioner has shown a persuasive motivation to pursue the claimed dose range of epinephrine (i.e., “between about 0.1 mg and about 2.4 mg”). PO Resp. 39–40. First, Patent Owner argues that “Petitioner fails to point to any art indicating that a single intranasal dose of epinephrine below 3 mg could elicit a sought-after biological or medicinal response in any mammal.” *Id.* at 41 (citing Ex. 2013 (Kaliner Decl.) ¶ 180; Ex. 2014 (Wyse Decl.) ¶ 145). Patent Owner asserts that “the epinephrine doses that Potta reported as showing increased plasma epinephrine concentrations when delivered intranasally were 6 mg,” and Srisawat “reported that for all intranasal doses tested **under** 5 mg (including 0.3, 0.6, 1.25, and 2.5 mg), there was no plasma epinephrine concentration and therefore no sought-after biological or medicinal response.” *Id.* at 40–41 (citing Ex. 1004 (Potta), Ex. 8; Ex. 2013 (Kaliner Decl.) ¶ 179; Ex. 2014 (Wyse Decl.) ¶ 145; Ex. 1006 (Srisawat), 4, Fig. 1).

Patent Owner's arguments are unavailing because claim 1 does not recite any limitation requiring a biological or medicinal response. Additionally, we agree with Petitioner that "to be motivated to pursue lower doses, efficacy at those doses need not be explicit in the art," and as discussed above, Petitioner demonstrates sufficient motivation to combine the teachings of Potta, Maggio, and Munjal to pursue formulations falling within the claimed dose range. Reply 17.

Patent Owner also argues that because Srisawat instructs that "the study should be repeated in a larger sample size and at the IN doses of 5 mg or higher to confirm the reproducibility of these findings and to obtain the optimal IN doses," Petitioner's argument that a person of ordinary skill in the art would have been motivated to "optimize" at 3 mg or lower lacks credibility. PO Resp. 42 (quoting Ex. 1006 (Srisawat), 5) (emphasis omitted).

This argument is unavailing. As Petitioner aptly explains, Srisawat's 5 mg intranasal formulation already demonstrated comparable absorption to the 0.3 mg intramuscular dose. Reply 19; Ex. 1006 (Srisawat), 1. Accordingly, a person of ordinary skill in the art looking to optimize for rapid onset would not add DDM to the 5 mg dose, given the risk of adverse effects with increased absorption. Reply 19; Ex. 1082 (Dyar Reply Decl.) ¶ 117; Ex. 1083 (Smyth Reply Decl.) ¶ 145. Instead, a person of ordinary skill in the art would have reasonably expected the addition of an absorption enhancer such as DDM to increase bioavailability, and thus would have started with a lower dose formulation. *See, e.g.*, Pet. 45; Ex. 1002 (Smyth Decl.) ¶ 128; Ex. 1005 (Maggio) ¶ 360; Ex. 1020 (Munjal), 5–6; Reply 19; Ex. 1083 (Smyth Reply Decl.) ¶¶ 105–06, 144–45; Ex. 1082 (Dyar Reply

Decl.) ¶ 117. We agree with Petitioner that “[i]n context, Srisawat’s instruction does not teach away considering the motivation to seek rapid absorption and lower doses.” Reply 19; *see also, e.g.*, Ex. 1003 (Weinstein Decl.) ¶¶ 100, 108, 137, 316–24, 336, 343; Ex. 1083 (Smyth Reply Decl.) ¶ 178 (“Skilled artisans would not understand [Srisawat’s suggestion] to be an instruction to dose at 5 mg or higher regardless of the pharmaceutical formulation.”).

Patent Owner additionally argues that because Srisawat shows “no response at all from epinephrine doses less than 5 mg administered intranasally,” there is no motivation “to use non-effective doses” and no reasonable expectation of success that lower doses “would actually work.” PO Resp. 42; Sur-reply 13.¹⁵ As to motivation, Patent Owner’s argument fails to account for the expected increase in bioavailability of lower doses resulting from the addition of DDM, as discussed above. As to reasonable expectation of success, Patent Owner’s argument is inapplicable, because the required success is defined by what is claimed, *see Intelligent Bio-Sys., Inc.*, 821 F.3d at 1367, and claim 1 does not recite any limitation requiring that the claimed formulations “would actually work.”

Patent Owner also argues that “[g]iven that neither of Petitioner’s experts were able to articulate what the maximum tolerable dose of epinephrine was or what the optimum dose was when delivered intranasally

¹⁵ Although we cite arguments from Patent Owner’s Sur-reply that fall under a heading relating to claim 3, Patent Owner applies its arguments made for claim 3 to claim 1. *See* Sur-reply 21 (“[W]ith respect to obviousness, the arguments for claim 1 of the ’414 patent regarding lack of motivation and reasonable expectation of success are the same as previously described for claim 3.”).

(or even transmucosally), it is not credible for them to claim that POSAs would have been motivated to formulate in the dosage range claimed by the '414 patent.” PO Resp. 42 (citing Ex. 2017 (Weinstein depo.), 101:13–22; Ex. 2018 (Smyth depo.), 35:8–16); *see also id.* at 40 (arguing that Dr. Weinstein was unable to identify a maximum or “optimum” dose of epinephrine, citing Ex. 2017 (Weinstein depo.), 118:2–6, 118:8–119:1).

Patent Owner does not adequately explain how these arguments undermine Petitioner’s showing. *See, e.g.*, Reply 17 (Petitioner explaining that Dr. Weinstein’s testimony (1) does not address whether the maximum dose was known, and (2) was explaining that “optimum” (a word not in claims) could mean different things in different contexts). Patent Owner’s focus on the maximum tolerable dose is inapt, particularly given epinephrine’s narrow therapeutic window, where efficacious doses are close to toxic doses, and the record shows that dose-dependent side effects occur at a variety of doses. *See* Pet. 21–22; Ex. 1041 (Simons 2006), Fig. 4. As discussed above, the record also shows that dose-dependent side effects would be expected to worsen when using an absorption enhancer that increases the speed and extent of drug uptake.¹⁶ *See also* Pet. 6; Ex. 1002

¹⁶ In Sur-reply, Patent Owner for the first time attacks two motivations Petitioner proffered in the Petition, i.e., (1) a motivation to pursue a lowered dose based on concern about epinephrine’s side effects; and (2) a motivation to optimize Potta’s formulations to achieve “rapid onset.” *See* Sur-reply 14, 15. Both of these motivations were fully argued in the Petition (*see, e.g.*, Pet. 6, 44, 45) and recognized in the Institution Decision (*see* Dec. 31), yet Patent Owner did not dispute these motivations in its Patent Owner Response. *See, e.g.*, Reply 18 (arguing that Patent Owner ignored Petitioner’s motivation arguments relating to, among other things, side effects and achieving rapid response). It is too late for Patent Owner to do so for the first time in Sur-reply. *See, e.g.*, Dec. 45 (“The Board will deem

(Smyth Decl.) ¶ 128; Ex. 2013 (Kaliner Decl.) ¶ 231 (Patent Owner’s declarant Dr. Kaliner testifying that “more rapid administration of epinephrine . . . may increase the risk of serious adverse effects such as ventricular arrhythmias, pulmonary edema, intracranial hemorrhage, and death”).

To the extent Patent Owner is arguing that a person of ordinary skill in the art would have stuck with the 5 and 6 mg doses disclosed in Potta and Srisawat because these doses “were well tolerated in patients” (*see* PO Resp. 42), we disagree that this undercuts Petitioner’s showing. This argument again fails to account for Petitioner’s arguments regarding a skilled artisan’s reason to pursue a lower-dose, rapid onset intranasal formulation that included an absorption enhancer such as DDM. *See, e.g.*, Pet. 45. As discussed above, we find that Petitioner sufficiently establishes that a person of ordinary skill in the art would have been motivated to lower the dose from 5 or 6 mg to compensate for the expected increase in bioavailability due to the addition of a permeation enhancer. *See id.*

To the extent Patent Owner argues persons of ordinary skill in the art would have used higher doses because that is what others were investigating, *e.g.*, for sublingual and inhalative administration (PO Resp. 42–43), we

waived any issue not raised in a timely response to the Petition.”); Paper 13 (Scheduling Order), 9 (“Patent Owner is cautioned that any arguments not raised in the response may be deemed waived.”). To the extent Patent Owner contends that its “rapid onset” arguments were provoked by Dr. Dyar’s discussion of how quickly IM injections onset (*see* Sur-reply 15 (citing Ex. 1082 (Dyar Reply Decl.) ¶ 54), that exact same information appears in the declarations submitted with the Petition. *Compare* Ex. 1082 (Dyar Reply Decl.) ¶ 54, *with* Ex. 1002 (Smyth Decl.) ¶ 84 and Ex. 1003 (Weinstein Decl.) ¶ 87.

agree with Petitioner that higher doses used in other delivery forms are less relevant. Reply 18. The record indicates that “drugs delivered by different routes of administration have different bioavailability,” and skilled artisans would have been aware that “doses must be adjusted for those routes of administration.” Ex. 1082 (Dyar Reply Decl.) ¶¶ 115, 116, 117; Ex. 1080 (Kaliner depo.), 144:20–145:8 (testifying that dosing for sublingual dosage form doesn’t necessarily translate to intranasal dosing); Ex. 1083 (Smyth Reply Decl.) ¶¶ 140–44, 179; Ex. 1006 (Srisawat), 4 (recognizing that higher dose needed for intranasal administration as compared to intramuscular administration, given that the drug must diffuse across the nasal epithelial barrier to reach the systemic circulation).

(2) *Patent Owner’s Arguments Regarding DDM*

Patent Owner next makes several arguments questioning whether Petitioner has shown a persuasive motivation to optimize Potta’s formulations by adding DDM, and a reasonable expectation of success in doing so. PO Resp. 43–51. First, Patent Owner argues that because Potta reports that bioavailability for its 6 mg “hydro-alcoholic” intranasal formulations “is as good or better than that of the intramuscular injection,” a person of ordinary skill in the art would have had no need to “optimize” these formulations. PO Resp. 44 (quoting Ex. 1004 (Potta) ¶ 269) (emphasis omitted). Patent Owner additionally argues that because Potta discloses benefits of these formulations, a person of ordinary skill in the art would have been motivated to use these hydro-alcoholic formulations, not aqueous formulations as recited in claim 1. *Id.* at 51 (citing Ex. 2013 (Kaliner Decl.) ¶ 128; Ex. 2014 (Wyse Decl.) ¶ 159; Ex. 1004 (Potta) ¶¶ 267–68 (disclosing

that subjects receiving the hydroalcoholic formulations had higher $C_{\max}$ versus those receiving the aqueous formulation)).

These arguments are unavailing because they do not address Petitioner's showing that a person of ordinary skill in the art would have been motivated to move away from Potta's hydro-alcoholic formulations due to potential irritation of the nasal mucosa. *See* Pet. 45; Ex. 1002 (Smyth Decl.) ¶ 389; Ex. 1003 (Weinstein Decl.) ¶ 342; Ex. 1047 (Kulkarni), 4 (“[N]on-aqueous nasal spray formulations are suspected to have increased levels of safety risk.”). Moreover, even if Potta's hydro-alcoholic formulations (or spray-dry formulations, *see* PO Resp. 51) were viable options, the formulations Petitioner asserts a person of ordinary skill in the art would have pursued need not be the best option, merely a suitable option from which the prior art did not teach away. *See Par Pharm., Inc.*, 773 F.3d at 1197–98.

Patent Owner next argues that “optimizing” a drug formulation does not necessarily mean using permeation enhancers, given that the fewer inactive ingredients added to a formulation, the better (e.g., in terms of ensuring product stability and reducing cost). PO Resp. 44–45 (citing Ex. 2014 (Wyse Decl.) ¶¶ 23, 31, 34, 40, 150, 152; Ex. 2016 (Smyth Decl. in IPR2017-00807) ¶ 42). Patent Owner has not demonstrated that this general concern is sufficiently strong here to undercut Petitioner's arguments, especially because Patent Owner has not identified any specific stability or cost concerns regarding Petitioner's proposed formulations. Additionally, even if there were ways to formulate without an absorption enhancer, this is not decisive, because the law does not require that an obvious formulation be

the only, or even the best, choice. *See* Reply 20; *Par Pharm., Inc.*, 773 F.3d at 1197–98.

Patent Owner relatedly argues that in addition to using permeation enhancers, many other approaches to improving a formulation’s bioavailability existed, such as using a spray-dried power, nanoparticles, or liposomes. PO Resp. 45 (citing Ex. 2014 (Wyse Decl.) ¶¶ 151; Ex. 2013 (Kaliner Decl.) ¶ 194; Ex. 1004 (Potta) ¶¶ 261–64). Here again, even if there were other suitable ways to optimize Potta’s formulations, we find that Petitioner has set forth one suitable approach and a persuasive rationale for adopting it, and, on this record, there is insufficient evidence to establish any teaching away from that approach.

Patent Owner next contends that even if a skilled artisan would have been motivated to optimize Potta’s formulations using a permeation enhancer, it is hindsight to say that they would have selected DDM, including because: (1) Potta’s formulation already included several ingredients that could function as a permeation enhancer (edetate disodium dehydrate, BKC, and ethanol); (2) there are hundreds of other potential permeation enhancers; (3) DDM had not been used in any FDA-approved medication and was not on the FDA’s “inactive ingredients list”; and (4) other permeation enhancers were preferred for use with epinephrine. PO Resp. 46–49 (citing, e.g., Ex. 2013 (Kaliner Decl.) ¶¶ 48, 129, 195–202, 217; Ex. 2014 (Wyse Decl.) ¶¶ 36–37, 39, 44, 45, 152–53, 156; Ex. 2018 (Smyth depo.), 64:10–65:16, 112:6–15).¹⁷

¹⁷ For the first time in Sur-reply, Patent Owner characterizes the Petition’s arguments regarding motivation to choose DDM as applying an “obvious to try” rationale. Sur-reply 16–17. The Petition does not indicate that it is

First, Patent Owner’s declarant Dr. Kaliner confirmed both that DDM was known to speed absorption, and that skilled artisans would have been motivated to speed absorption for an epinephrine nasal spray. Ex. 1080 (Kaliner depo.), 163:5–14; Reply 21–22. This testimony is consistent with Petitioner’s asserted motivation to use DDM to achieve rapid onset. *See, e.g.*, Pet. 45. Second, the existence of permeation enhancers other than DDM is not decisive. Petitioner demonstrates a motivation to use DDM, and we see no showing on this record that the prior art taught away from DDM. *See Par Pharm., Inc.*, 773 F.3d at 1197–98. The identification of other permeation enhancers as “preferred” or “standard” does not constitute a teaching away. *Galderma Labs., L.P. v. Tolmar, Inc.*, 737 F.3d 731, 739 (Fed. Cir. 2013) (“A teaching that a composition may be optimal or standard does not criticize, discredit, or otherwise discourage investigation into other compositions.”).

As to DDM in particular, Patent Owner’s argument that “[s]ome articles even questioned the viability of dodecylmaltoside as an inactive ingredient due to toxicity concerns” is not well-supported. PO Resp. 47 (citing Ex. 2026 (Vllasaliu), 1, 8–11; Ex. 2013 (Kaliner Decl.) ¶ 166; Ex. 2014 (Weiss Decl.) ¶ 154). Although Patent Owner and its declarants reference “articles” (plural), they cite one such article, Vllasaliu (Ex. 2026). *See id.*; *see also* Ex. 1081 (Wyse depo.), 133:17–134:14 (Patent Owner’s declarant Dr. Wyse could not recall whether additional articles suggested

applying this legal theory. Moreover, Patent Owner did not raise its “obvious to try” argument in the Patent Owner Response, and thus Patent Owner forfeited this argument. *See, e.g.*, Dec. 45; Paper 13 (Scheduling Order), 9.

that DDM is toxic). Vllasaliu reports results from an *in vitro* assay using Calu-3 cells. *See* Ex. 2026 (Vllasaliu), Abstr.; Ex. 1083 (Smyth Reply Decl.) ¶ 132. The relevance of this cell model for toxicity assessment was criticized in a subsequent paper, which stated: “[a]ttempts to use this same cell based model for toxicity assessment is not predictive because of gross difference between the *in vitro* and *in situ* cell environments.” Ex. 1016 (Maggio 2014), 5; Ex. 1083 (Smyth Reply Decl.) ¶ 132. Moreover, we agree with Dr. Smyth that skilled artisans would have placed more importance on Munjal’s clinical trial results over Vllasaliu’s *in vitro* data. Ex. 1083 (Smyth Reply Decl.) ¶ 132. As Dr. Smyth explains, Munjal reported that intranasal administration of a formulation comprising DDM was safe and tolerable, and all adverse events “were mild in severity.” *Id.* (quoting Ex. 1020 (Munjal), 7); *see also* Pet. 24 (“Maggio disclosed that Intravail®, such as DDM, was safe, non-toxic, and non-irritating, in contrast to other absorption enhancement agents that cause damage to the nasal mucosa.”); Ex. 1005 (Maggio) ¶ 73; Ex. 1015 (Maggio 2013), 2–3; Ex. 1016 (Maggio 2014), 5–6.

Patent Owner attempts to distinguish prior art that teaches or suggests the use of DDM with epinephrine. PO Resp. 48–49. Specifically, Kee teaches an epinephrine ophthalmic formulation comprising DDM, BKC, and a viscosity agent (hydroxypropylmethylcellulose (“HMPC”)). Ex. 1021 (Kee), 9:15–38, 8:23–33. Kee reports that when this formulation was administered to rabbits, it permeated “far better through the cornea” than a formulation having a 10-fold higher epinephrine concentration that lacked DDM and HMPC. *Id.* Patent Owner asserts that a skilled artisan would not have viewed Kee’s results “as relevant to, or predictive of, the problem of

intranasal delivery of epinephrine,” given differences such as the use of a viscosity agent, the structure of the cornea versus that of the nasal mucosa (including mucus, cilia, etc.), and the topical/local response measured in Kee (change in pupil diameter), as opposed to a systemic effect sought here (increased plasma epinephrine concentration). PO Resp. 48 (citing Ex. 2013 (Kaliner Decl.) ¶ 198; Ex. 2014 (Wyse Decl.) ¶ 155).

Despite the differences in formulation and measured or desired effect, however, we find that a person of ordinary skill in the art would have viewed Kee as demonstrating the compatibility of DDM and epinephrine for transmucosal delivery. Ex. 1002 (Smyth Decl.) ¶ 377; Ex. 1003 (Weinstein Decl.) ¶ 329. Dr. Kaliner posits that “it is unclear what effect dodecylmaltoside would have had on epinephrine in the absence of HPMC.” Ex. 2013 (Kaliner Decl.) ¶ 198. But Dr. Kaliner admits, “I am not an expert in the formulation of intranasal drugs,” and Patent Owner’s formulation expert, Dr. Wyse, did not raise the same question. *Id.* ¶ 33; *see also* Ex. 2014 (Wyse Decl.) ¶ 155 (addressing Kee).

Kee teaches that HPMC, a viscosity agent, allows the drug to be retained in the eye for a longer period of time, “thus allowing the enhancers more time to facilitate drug transport through the cornea.” Ex. 1021, 2:27–35, 5:37–49. We have not been directed to anything in the record indicating that HPMC is necessary for DDM to effect transmucosal delivery of epinephrine or any other drug. Indeed, Kee teaches and claims example formulations using DDM with a drug but no HPMC or other viscosity enhancer. Ex. 1021, claims 1–3, 11:36–48 (Example 11); Ex. 1002 (Smyth Decl.) ¶¶ 207–08. Thus, without identifying a specific reason why HPMC might have been necessary to effect permeation of epinephrine into the

cornea, Dr. Kaliner’s speculation does not adequately suggest that a person of ordinary skill in the art would have discounted Kee’s teachings.

Dr. Kaliner further asserts that “the differences in the structure of the cornea versus the nose could impact the permeation abilities of dodecylmaltoside.” Ex. 2013 (Kaliner Decl.) ¶ 198. But again, Dr. Kaliner fails to provide any specifics here, e.g., by contrasting how the different structures of the cornea and nose could differentially impact the ability of DDM to enhance epinephrine’s movement across the mucosa. *See id.* Drs. Kaliner and Wyse identify the mucus barrier lining the mucosal epithelium of tissues as one of the major challenges to transmucosal drug delivery, but a mucus barrier exists in the eye as well. Ex. 2013 (Kaliner Decl.) ¶ 35; Ex. 2014 (Wyse Decl.) ¶ 24; Ex. 2024 (Leal), 1.

Patent Owner also seeks to distinguish Hill (Ex. 1022). PO Resp. 48–49. Hill teaches buccal, lingual, and sublingual epinephrine tablets for the treatment of anaphylaxis, where the tablets can include absorption enhancers such as DDM. Ex. 1022 (Hill), code (57), ¶ 53. Patent Owner criticizes Hill for lacking “actual data or examples.” PO Resp. 48–49 (citing Ex. 2013 (Kaliner Decl.) ¶ 199¹⁸; Ex. 2014 (Wyse Decl.) ¶ 155). We agree with Petitioner’s declarant Dr. Smyth, however, that in the context of the prior art as a whole—including other references teaching or suggesting the use of DDM with epinephrine and other drugs in transmucosal and intranasal formulations—Hill would have provided skilled artisans with additional

¹⁸ Patent Owner cites Ex. 2013 at paragraph 198, but that paragraph relates to Kee. PO Resp. 48–49. We assume that Patent Owner intended to cite paragraph 199, which relates to Hill.

motivation to combine DDM with epinephrine in an intranasal formulation. Ex. 1083 (Smyth Reply Decl.) ¶ 160.

Patent Owner also seeks to discount Maggio and Munjal because they teach the use of DDM with opioids and sumatriptan, not with epinephrine. PO Resp. 49–50 (citing, e.g., Ex. 2014 (Wyse Decl.) ¶ 157; Ex. 2013 (Kaliner Decl.) ¶¶ 200–01; Ex. 2017 (Weinstein depo.), 149:10–18; Ex. 1064 (Pires), 17). Patent Owner correctly notes that epinephrine, sumatriptan, and opioids have different structures and indications. *See id.* at 48–49. But Patent Owner’s narrow focus on the different structures and indications overlooks that the prior art—including Maggio itself—in fact taught combining epinephrine and DDM. *See* Ex. 1005 (Maggio) ¶ 356. On the complete trial record, we find that these teachings—together with Maggio and Munjal’s teachings that DDM increases uptake and bioavailability (albeit demonstrated with other molecules)—sufficiently support a motivation to combine Potta’s intranasal epinephrine formulations with DDM.

Patent Owner also ignores that the prior art taught that “permeation effectiveness of certain alkylsaccharides is largely a function of the molecular weight of the drug with greater bioavailabilities observed with lower molecular weight drugs.” Pet. 24; *see also id.* at 25; Ex. 1015 (Maggio 2013), 6; Reply 22; Ex. 1002 (Smyth Decl.) ¶¶ 125–26, 227–28; Ex. 1025 (Cunningham), Abstr., 1–2, 4; Ex. 1083 (Smyth Reply Decl.) ¶¶ 97–102, 162; Ex. 1082 (Dyar Reply Decl.) ¶ 124. The prior art teaches that “[f]or small molecule drugs (i.e., MW<1,000 Daltons) nasal bioavailability in the presence of alkylsaccharide absorption enhancers is predominantly a function of molecular weight and is expected to be high.”

Ex. 1025 (Cunningham), 2. DDM increased the absorption and rapid onset of action of sumatriptan (molecular weight of 295.4 g/mol) and hydromorphone (molecular weight of 285.3 g/mol). *See* Ex. 1082 (Dyar Reply Decl.) ¶¶ 123–24; Ex. 1020 (Munjai) (sumatriptan), 5–10; Ex. 1025 (Cunningham) (hydromorphone), 2, 4). These teachings support Petitioner’s contention that DDM would reasonably be expected to increase the absorption and rapid onset of action of epinephrine (molecular weight of 183.2 g/mol), which is smaller than sumatriptan and hydromorphone, as well. Ex. 1082 (Dyar Reply Decl.) ¶ 124. And indeed, Kee demonstrated as much in an ocular formulation. Ex. 1021 (Kee), 9:15–38.

Although Patent Owner’s Response does not directly address Petitioner’s arguments regarding the importance of molecular weight on DDM’s efficacy, Patent Owner does assert that “the effects of dodecylmaltoside are (a) largely unpredictable, and (b) can have no absorption enhancing effects on certain molecules.” PO Resp. 50 (citing Ex. 2028 (Gradauer), 1, 5; Ex. 2014 (Wyse Decl.) ¶ 158; Ex. 2013 (Kaliner Decl.) ¶ 202; Ex. 2029 (Crystal 556); Ex. 2033 (Crystal 666)). In support, Patent Owner cites Gradauer, which describes a study designed “to provide a better understanding of the molecular interaction of absorption enhancers with the intestine.” Ex. 2028 (Gradauer), Abstr. In particular, to determine the “extent to which the action of DDM is apportioned between the para- and transcellular routes” in Caco-2 cells, the study “examin[ed] the transport of relevant marker compounds ([³H]-mannitol and [³H]-propranolol, respectively).” *Id.* The results showed that “[i]n the case of [³H]-mannitol, a robust permeation enhancement was achieved with 0.5 mM DDM (~6-fold), whereas little effect was seen on the permeation of [³H]-propranolol.”

Id. Based on Gradauer’s results showing little permeation of the [³H]-propranolol marker, Patent Owner’s declarants conclude that DDM does not have an absorption-enhancing effect on every molecule. Ex. 2013 (Kaliner Decl.) ¶ 202; Ex. 2014 (Wyse Decl.) ¶ 158.

On this record, this study’s relevance to a skilled artisan is unclear. *See* Ex. 1083 (Smyth Reply Decl.) ¶ 133 (“[T]hat Gradauer 2017 showed different ‘effects’ on the transport markers, mannitol and propranolol, does not relate to the effects that dodecyl maltoside would have on small molecules in pharmaceutical formulations.”). And even if Gradauer demonstrates that DDM transported one small molecule or marker ([³H]-mannitol) but not another ([³H]-propranolol) in an *in vitro* intestinal cell model, we find more relevant other art such as Kee, which demonstrates that DDM transported epinephrine across the ocular mucosal membrane in the rabbit. *See* Ex. 1021 (Kee), 9:15–38.

Patent Owner’s declarant Dr. Wyse also asserts that “because DDM and other alkyl saccharides work via a ‘multimodal’ mechanism (i.e., they act both via para- and trans-cellular mechanisms), predictability of their behavior with any given molecule is difficult.” Ex. 2014 (Wyse Decl.) ¶¶ 47, 158 (citing Ex. 2030 (Danielsen 2018)). But even if the exact cellular transport route differs across molecules, this does not necessarily mean that whether DDM will transport a given molecule is unpredictable. *See* Ex. 1083 (Smyth Reply Decl.) ¶ 34.

Patent Owner’s citation to Exhibits 2029 and 2033, which are two patent applications to Crystal, is unavailing because neither Patent Owner nor its declarants cite any particular portion of these exhibits, or otherwise explain their relevance. *See* PO Resp. 50; Ex. 2014 (Wyse Decl.) ¶¶ 47,

158; Ex. 2013 (Kaliner Decl.) ¶ 202. Accordingly, we decline to consider these exhibits. *See* Trial Practice Guide 66 (patent owner response should “explain the relevance” of evidence sought to be relied upon).

In sum, we find that Patent Owner’s arguments and evidence do not trump the prior art showing that epinephrine and DDM actually moved together across rabbit ocular epithelium. *See* Ex. 1021 (Kee), 9:15–38. We determine that this more relevant evidence, together with the general knowledge that for “small molecule drugs (i.e., MW<1,000 Daltons) nasal bioavailability in the presence of alkylsaccharide absorption enhancers is predominantly a function of molecular weight and is expected to be high” (Ex. 1025 (Cunningham), 2), is sufficient on this record to demonstrate a motivation to select DDM with a reasonable expectation of success.

(3) Patent Owner’s Arguments Regarding Long-Felt Need

Patent Owner argues that the claimed subject matter satisfies a long-felt, unresolved need. Patent Owner does not clearly identify the contours of the alleged long-felt, unresolved need. As best we can discern, Patent Owner appears to be arguing that there was a long-felt, unresolved need for an alternative to the treatment of choice for type-1 hypersensitivity reactions, which is an intramuscular injection of epinephrine. *See, e.g.*, PO Resp. 37–38, 56–58 (discussing how skilled artisans have unsuccessfully searched for alternatives to epinephrine intramuscular injections).

We agree with Patent Owner that the record demonstrates disadvantages associated with intramuscular injection, and a search for alternative delivery routes (including via intranasal delivery). *See, e.g.*, PO Resp. 56–59; Pet. 20; Ex. 1010 (Fleming) ¶ 6 (“[T]here exists a need in

the market for more convenient, easy delivery of correct dosage form that does not require prior training in the use of the device, and increase compliance in persons prone to anaphylaxis.”); Ex. 1006 (Srisawat), 2 (“[A]lternatives to epinephrine auto-injectors have been investigated, e.g., epinephrine administered via inhalation, sublingual, or intranasal (IN) routes.”); Ex. 1004 (Potta) ¶ 6 (“While there are various epinephrine formulations currently available, there remains a need in the art for a quick-onset spray formulation.”); Ex. 1022 (Hill) ¶ 7 (“there exists a need in the art for additional methods of immediately administering epinephrine to a person undergoing anaphylaxis wherein the epinephrine can be administered in a series of sequential dosage forms which do not rely solely on epinephrine injectors”).

We find, however, that Patent Owner has not adequately demonstrated that any long-felt need for an alternative to intramuscular injection remained unmet until the claimed subject matter satisfied the need. *See, e.g., In re Cavanagh*, 436 F.2d 491, 496 (CCPA 1971) (“even if we accept that there was a problem, it was still incumbent upon appellant . . . to bring forward evidence of his satisfaction of the need”). First, the record shows, and Patent Owner acknowledges, that persons of ordinary skill in the art were exploring a variety of epinephrine delivery routes, including sublingual, lingual, buccal, inhalative, and intranasal delivery. *See, e.g.,* POREsp. 57. Patent Owner asserts, however, that “none of these efforts have resulted in a commercial product.” We are not aware of any legal requirement that there

must be a “commercial product” to demonstrate prior satisfaction of a long-felt need.¹⁹

We agree with Petitioner that at least Potta and Fleming’s intranasal formulations satisfied any need for an alternative to epinephrine intramuscular injections. *See* Ex. 1003 (Weinstein Decl.) ¶ 456; Reply 25; Ex. 1083 (Smyth Reply Decl.) ¶ 190; Ex. 1002 (Smyth Decl.) ¶ 93; Ex. 1004 (Potta) ¶ 243 (Table 1), claim 21; Ex. 1010 (Fleming) ¶¶ 12, 15, 18, 31–32. Patent Owner does not clearly dispute Petitioner’s contentions that the Potta and Fleming formulations satisfy any long-felt need, and indeed, Patent Owner and its declarants acknowledge that Potta’s 6 mg intranasal formulations achieved a biological response and were well-tolerated. *See* PO Resp. 40, 42, 44; Ex. 2013 (Kaliner Decl.) ¶¶ 32, 179, 181, 193; Ex. 2014 (Wyse Decl.) ¶¶ 145, 147; Ex. 1004 (Potta), Example 8.

Patent Owner asserts that the challenged patent “for the first time” provides an alternative to the intramuscular administration using “doses below 3 mg with an intranasal route.” PO Resp. 58 (citing Ex. 2013 (Kaliner Decl.) ¶ 232). Thus, to the extent Patent Owner seeks to discount

¹⁹ In any event, we observe that Patent Owner’s declarant Dr. Kaliner discusses a marketed product, namely “an inhaled epinephrine product (INFECTOKRUPP® Inhal) . . . marketed in Germany for the treatment of allergic reactions of the upper airways.” Ex. 2013 (Kaliner Decl.) ¶ 29 (citing Ex. 1050 (Breuer), 1–2), *see also id.* ¶ 184. Potta also mentions previously-available epinephrine nasal and aerosol sprays, and a commercially available oral inhalation solution of a racemate of epinephrine. Ex. 1004 (Potta) ¶ 4. We also observe that to the extent a commercial product is required to satisfy a long-felt need, Patent Owner has not pointed to a commercial product with sufficient nexus to the challenged claims that satisfies a long-felt need.

Potta or Fleming's formulations as solving the long-felt need on the basis that they are higher doses than recited in the claims, we agree with Petitioner that Patent Owner has identified no nexus between a long-felt but unmet need and the claimed doses. *See* Reply 25–26; *see also Iron Grip Barbell Co. v. USA Sports, Inc.*, 392 F.3d 1317, 1324–25 (Fed. Cir. 2004) (affirming summary judgment of obviousness where patentee failed to prove nexus). And to the extent there was a long-felt need for an intranasal epinephrine formulation below 3 mg, we agree with Petitioner that the anticipating formulations in Potta discussed above (in Section II.D.2.a) satisfy that need. *See* Reply 25; Ex. 1083 (Smyth Reply Decl.) ¶ 190 (Petitioner's declarant Dr. Smyth opining that the anticipatory 1% formulations of Potta Table 1 or claim 21 satisfy any long-felt but unmet need for a low-dose epinephrine intranasal formulation).

For the above reasons, we determine that Patent Owner has not demonstrated that the claimed subject matter satisfied a long-felt, unmet need.

(4) *Patent Owner's Arguments Regarding
Unexpected Results*

Patent Owner asserts that around the time of the challenged patent, skilled artisans “were investigating several different methods of transmucosal delivery of epinephrine, and using much higher doses than intramuscular injection to achieve a systemic response.” PO Resp. 59 (citing, e.g., Ex. 1004 (Potta) ¶¶ 265–69 (6 mg intranasal spray); Ex. 1006 (Srisawat), 4, Fig. 1 (5 mg intranasal spray); Ex. 1051 (Rawas Qalaji), 2, 4 (10, 20, 40 mg sublingual tablets); Ex. 2032 (Levitt), 1 (20 mg sublingual tablets); Ex. 1050 (Breuer), 1–2 (4 and 8 mg inhalative doses); Ex. 2013

(Kaliner Decl.) ¶ 233). Patent Owner further asserts that Srisawat showed “no systemic absorption of epinephrine when administered intranasally at doses of 2.5 mg and lower.” *Id.* (citing, e.g., Ex. 1006 (Srisawat), 3, Fig. 1). Patent Owner asserts that its “discoveries with respect to lower doses of epinephrine were entirely unexpected.” *Id.*

We are not persuaded by Patent Owner’s arguments. Purported unexpected results “must be shown to be unexpected compared with the closest prior art.” *In re Baxter Travenol Labs.*, 952 F.2d 388, 392 (Fed. Cir. 1991). We agree with Petitioner that Patent Owner’s “unexpected results argument identifies no comparison to the closest prior art or a nexus to the claimed invention.” Reply 26; *see also* Ex. 1003 (Weinstein Decl.) ¶ 455 (opining that that “there is no evidence that any of the claimed ranges are ‘critical’ or achieve ‘unexpected results,’ when compared, for example, to the ranges claimed in Potta”). We are also not persuaded that the dosages used for other routes of administration are relevant to whether the claimed dosage for intranasal administration are “surprising” or “unexpected.” The record indicates that “drugs delivered by different routes of administration have different bioavailability,” and skilled artisans would have been aware that “doses must be adjusted for those routes of administration.” Ex. 1082 (Dyar Reply Decl.) ¶¶ 115, 116, 117; Ex. 1080 (Kaliner depo.), 144:20–145:8 (testifying that dosing for sublingual dosage form doesn’t necessarily translate to intranasal dosing); Ex. 1083 (Smyth Reply Decl.) ¶¶ 140–43.

Patent Owner further contends that alkylsaccharides “had not been adopted for use before the priority date,” and that significant toxicity of alkylsaccharides to airway epithelial cells had been reported. PO Resp. 59 (citing, e.g., Ex. 2013 (Kaliner Decl.) ¶¶ 50–53, 235; Ex. 2026 (Vllasaliu), 1,

8–11). Patent Owner asserts: “Thus, it was unexpected that the inventors of the ’414 patent were able to combine dodecylmaltoside with epinephrine administered intranasally and arrive at the doses claimed in the ’414 patent.” *Id.* (citing Ex. 2013 (Kaliner Decl.) ¶ 235). This argument is unavailing. We agree with Petitioner that to the extent Patent Owner is arguing the claimed invention is unexpectedly nontoxic, prior art clinical administration of intranasal formulations comprising DDM performed after Vllasaliu demonstrated no toxicity concern. Reply 26; Ex. 1083 (Smyth Reply Decl.) ¶ 193; Ex. 1020 (Munjal), 10 (administration was “safe and well tolerated”).

For the above reasons, we determine that Patent Owner has not demonstrated unexpected results associated with the claimed subject matter.

(5) Conclusion on Claim 1

Considering all of the arguments and evidence presented on the full trial record, we determine that Petitioner has demonstrated by a preponderance of the evidence that claim 1 would have been unpatentable as obvious over Potta, Maggio, and Munjal. Pet. 48–51.

b) Dependent Claims 2, 7–9 and 11–17

Dependent claims 2, 7–9, and 11–17 depend directly or indirectly from claim 1. As to claims 2, 7–9, and 12–17, Petitioner argues that the cited prior art discloses the additional limitations introduced in each of these dependent claims, and supports its argument with citations to the prior art and testimony from its declarants. *See* Pet. 51 (claims 2, 12, 13, 15–17), 52 (claims 7–9), 53 (claim 14). Other than its arguments over the subject matter of claim 1 incorporated into these dependent claims, which we discussed above, Patent Owner proffers no arguments specific to these dependent claims (and thus has forfeited any arguments specific to these

claims). *See* PO Resp. 56; *In re NuVasive, Inc.*, 841 F.3d at 974; Paper 13, 9. Thus, based on our review of Petitioner's arguments and evidence relating to dependent claims 2, 7–9, and 12–17 (*see* Pet. 51 (claims 2, 12, 13, 15–17), 52 (claims 7–9), 53 (claim 14)), which we adopt as our own, and our analysis of claim 1 in Section II.E.2.a above, we conclude that Petitioner has proved by a preponderance of the evidence that claims 2, 7–9, and 12–17 are unpatentable as obvious over the combination of Potta, Maggio, and Munjal.

As to dependent claim 11, in our Institution Decision we expressed reservations regarding the sufficiency of Petitioner's arguments on this claim, particularly whether the Petition sufficiently articulates a reason why a person of ordinary skill in the art would have been motivated to use a combination of BKC and DDM as absorption enhancers in Potta's formulation. Dec. 39–40. At that time, we did not recognize that Potta expressly discloses using BKC as an absorption enhancer. However, and as discussed above in connection with our anticipation analysis of claim 11, post-institution, both parties assert that Potta in fact discloses BKC as an absorption enhancer. *See* Ex. 1004 (Potta) ¶ 20; Ex. 1083 (Smyth Reply Decl.) ¶ 88; Ex. 2014 (Wyse Decl.) ¶ 110; Ex. 1080 (Kaliner depo.), 174:18–175:22. Potta also discloses using amounts of permeation enhancers that encompass the claimed range, as well as using a combination of permeation enhancers. Ex. 1004 (Potta) ¶¶ 20, 85. Maggio also discloses using BKC as a permeation enhancer, using amounts that encompass the claimed amounts, and using a combination of permeation enhancers. Ex. 1005 (Maggio) ¶¶ 171, 189, 243; *see also* Ex. 1002 (Smyth Decl.) ¶ 470. Patent Owner does not specifically dispute Petitioner's obviousness arguments concerning claim 11. *See generally* PO Resp. In view of Potta

and Maggio's disclosures and the arguments of record, we conclude that Petitioner has proved by a preponderance of the evidence that claim 11 is unpatentable as obvious over the combination of Potta, Maggio, and Munjal. *See, e.g.*, Pet. 53; Reply 25; Ex. 1002 ¶¶ 467–76; Ex. 1083 ¶¶ 172–73, 175–76.

c) Dependent Claim 10

Pertinent to Petitioner's obviousness argument, claim 10 recites in relevant part: "about 0.005% (w/v) to about 2.5% (w/v) dodecyl maltoside." Ex. 1001, 73:17–18. Petitioner asserts that Maggio and Munjal teach amounts of DDM that overlap with or are encompassed by claim 10, and specifically teach using 0.18% and 0.20% DDM in an intranasal sumatriptan formulation. Pet. 53.

To show obviousness, however, it is not enough to merely show that the prior art includes references disclosing each limitation in a challenged claim. *See Unigene Labs., Inc. v. Apotex, Inc.*, 655 F.3d 1352, 1360 (Fed. Cir. 2011). In the Institution Decision, we questioned whether the Petition identifies a sufficient reason why a person of ordinary skill would have been motivated to use the amounts of DDM taught in Maggio and Munjal's sumatriptan formulations in Potta's epinephrine formulation. Dec. 39; *see also Metalcraft of Mayville, Inc. v. Toro Co.*, 848 F.3d 1358, 1366 (Fed. Cir. 2017) ("In determining whether there would have been a motivation to combine prior art references to arrive at the claimed invention, it is insufficient to simply conclude the combination would have been obvious without identifying any reason why a person of skill in the art would have made the combination."). Post-institution, Petitioner asserts that Patent Owner "has come forth with nothing and thus admits obviousness" of the

additional limitation of claim 10. Reply 25. This is an incorrect statement of law. Petitioner bears the burden of demonstrating unpatentability, and that burden remains with Petitioner throughout IPR proceedings. *See Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016); *Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015).

As we stated in the Institution Decision, although the Petition notes that both sumatriptan and epinephrine are small molecules (Pet. 53), this is too tenuous a reason to have motivated a skilled artisan to use the specific amounts of DDM (0.18% and 0.20%) used in the sumatriptan formulations in an epinephrine formulation. Dec. 38. On Reply, Petitioner additionally states that the “prior art taught using ~0.2% DDM with small molecules,” and cites additional paragraphs from Dr. Smyth’s Reply Declaration comparing the molecular weights of sumatriptan, hydromorphone, and epinephrine, and suggesting that given the “similar molecular weight,” a person of ordinary skill in the art would have been motivated to use the same amounts of DDM with epinephrine that were used with sumatriptan and hydromorphone. Reply 25; Ex. 1083 (Smyth Reply Decl.) ¶ 171. This more specific explanation and discussion of relative molecular weights is missing from the Petition’s arguments for claim 10. As stated in the Trial Practice Guide, “Petitioner may not submit new evidence or argument in reply that it could have presented earlier, e.g. to make out a prima facie case of unpatentability.” Trial Practice Guide, 73.

Accordingly, on the full trial record, we conclude that Petitioner has not proved by a preponderance of the evidence claim 10 is unpatentable as obvious over the combination of Potta, Maggio, and Munjal.²⁰

d) Dependent Claims 3–6 and 18–20

Petitioner does not attempt to demonstrate that the additional limitation recited in claim 3, namely “a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis,” is expressly recited in the asserted prior art. *See, e.g., Par Pharm., Inc.*, 773 F.3d at 1194 (“We first must determine whether [the patent challenger] carried its burden to prove that all claimed limitations are disclosed in the prior art.”) Instead, Petitioner asserts that combining Potta’s formulations with DDM would “result[] in a formulation that would inherently, at a minimum, ‘elicit a biological response,’ as well as a higher biological effect.” Pet. 52.

“[I]nherency may supply a missing claim limitation in an obviousness analysis,” but this is a “high standard.” *Par Pharm., Inc.*, 773 F.3d at 1194–95. Petitioner must establish that “the limitation at issue [is] necessarily . . .

²⁰ Above we find that Potta anticipates claim 10. *See* Section II.D.2.a. Although we recognize the maxim that “anticipation is the epitome of obviousness” (*see, e.g., In re McDaniel*, 293 F.3d 1379, 1385 (Fed. Cir. 2002)), here we find that Petitioner has not carried its burden of demonstrating that claim 10 is obvious. This outcome is a result of the Petitioner’s specific arguments. *See SAS Institute, Inc. v. Iancu*, 138 S. Ct. 1348, 1356 (2018) (indicating that the Petition guides the life of the trial). Claim 10 limits claims 1 and 7 by reciting specific amounts of DDM, BKC, and/or oleic acid, and combinations thereof. *See* Ex. 1001, 73:14–31. Petitioner’s claim 10 anticipation arguments are based Potta’s disclosures for BKC and oleic acid, but its obviousness arguments are based on Maggio and Munjal’s disclosures of DDM. *Compare* Pet. 39, *with id.* at 53.

present, or the natural result of the combination of elements explicitly disclosed by the prior art.” *Id.* at 1195–96.

The difference between Petitioner’s anticipation and obviousness arguments for claim 3 is that for claim 3, Petitioner relies on the addition of DDM to Potta’s formulations. *Compare* Pet. 35, *with id.* at 52. Thus, the operative question here is whether Potta’s formulations with DDM would inherently provide the recited “plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis.”

The Petition’s argument and evidence on this point is thin. Petitioner asserts that Maggio and Munjal “confirm[]” that Potta’s formulations with DDM would elicit a biological response, because these references show that DDM “provide[s] rapid and increased absorption.” Pet. 52 (citing Ex. 1002 (Smyth Decl.) ¶¶ 405–06, 418–19; Ex. 1003 (Weinstein Decl.) ¶¶ 353–54, 366–68). Thus, even if a person of ordinary skill in the art would have expected DDM to enhance the onset and bioavailability of epinephrine (*see, e.g.*, Ex. 1002 (Smyth Decl.) ¶ 405), the pertinent question is whether adding DDM to the cited Potta formulations would *necessarily* result in the claimed plasma epinephrine concentration. We are unconvinced that Maggio and Munjal answer this question, because Petitioner has not sufficiently demonstrated that what applied to phenylephrine and sumatriptan would have also applied to the same extent to Potta’s formulations. Indeed, the record reflects that “[e]ach drug is one particular case and, thus, the relationship between the drug characteristics, the strategies considered and the permeation rate is essential.” Ex. 1064 (Pires), 17.

Petitioner points to data in Srisawat and Potta regarding the administration of 5 mg and 6 mg intranasal epinephrine formulations,

respectively. Pet. 46. It is undisputed that these formulations—which lacked DDM—resulted in a plasma epinephrine concentration. *See, e.g.*, Ex. 1006 (Srisawat), Fig. 1; Ex. 1004 (Potta), Example 8; PO Resp. 52; Pet. 46. We are also persuaded that if DDM were added to Srisawat and Potta’s formulations, it would be possible to lower the dose from 5 mg and 6 mg and still likely achieve epinephrine absorption in the formulation recipients. *See, e.g.*, Pet. 50; Ex. 1002 (Smyth Decl.) ¶¶ 128–30, 393–396; Ex. 1020 (Munjal), 5–6, 9.

What is missing on this record, however, is the extent to which the 5 mg and 6 mg doses in Srisawat and Potta could be lowered due to the addition of DDM, and still *necessarily* result in the claimed plasma epinephrine concentration. Or more to the point—nothing on this record indicates that if DDM were added to the Potta formulations on which Petitioner relies, that they would *necessarily* result in “a plasma epinephrine concentration,” let alone one “that is efficacious for the treatment of anaphylaxis” as recited in claim 3.

The Petition relies on the “high standard” of inherency, which requires that Petitioner show “the limitation at issue [is] necessarily . . . present, or the natural result of the combination of elements explicitly disclosed by the prior art.” *Par Pharm., Inc.*, 773 F.3d at 1194–95; Pet. 52. This is not a case where a plasma epinephrine concentration is expected from administration of the cited Potta formulations with DDM. *Cf. Santarus, Inc. v. Par Pharm., Inc.*, 694 F.3d 1344, 1354 (Fed. Cir. 2012) (affirming obviousness where claim recited specific serum concentrations not disclosed in the prior art, but it was undisputed that those concentrations were expected in light of the claimed dosages). Rather, the record shows

that some epinephrine intranasal dosages (e.g., Srisawat’s 5 mg dosage and Potta’s 6 mg dosage) resulted in epinephrine absorption, while others did not (e.g., Srisawat’s 0.3, 0.6, 1.25, and 2.5 mg dosages). *See, e.g.*, Ex. 1006 (Srisawat), 1, Fig. 1; Ex. 1004 (Potta), Example 8; PO Resp. 41 n.6. Nothing on this record persuades us that the lower-dose Potta formulations, even with the “boost” likely provided by DDM, would—not possibly or probably, but *necessarily*—result in “a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis,” as recited in claim 3. *See Par Pharm., Inc.*, 773 F.3d at 1194–95.

For the above reasons, we determine on the full trial record that Petitioner has not demonstrated by a preponderance of the evidence that claim 3 would have been obvious over Potta, Maggio, and Munjal. Because claims 4–6 and 18 depend directly or indirectly from claim 3, and because Petitioner’s arguments for these claims do not rectify the deficiencies discussed above for claim 3, we similarly determine on the full trial record that Petitioner has not demonstrated by a preponderance of the evidence that claims 4–6 and 18 would have been obvious over Potta, Maggio, and Munjal.

3. *Alleged Obviousness Over Srisawat, Maggio, and Munjal*

Petitioner contends that claims 1–20 would have been obvious over Srisawat, Maggio, and Munjal. Pet. 54. Because we have already found claims 1, 2, and 7–17 unpatentable on other grounds, we limit our analysis of this ground to dependent claims 3–6 and 18–20. *SAS Inst. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner “is entitled to a final written decision addressing all of the claims it has challenged”); *Boston Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App’x 984, 990 (Fed. Cir. 2020)

(nonprecedential) (stating that the “Board need not address issues that are not necessary to the resolution of the proceeding,” e.g., “alternative arguments with respect to claims [the Board] found unpatentable on other grounds”).

As discussed above (*see supra* Section II.E.2.a), Petitioner contends that persons of ordinary skill in the art would have sought to optimize low dose intranasal epinephrine formulations. Applying this premise to its Srisawat-based arguments, Petitioner contends that skilled artisans, “understanding [the] side effects posed by higher [epinephrine] dosing and looking to optimize intranasal dosing (e.g., the 1.25 or 2.5 mg Srisawat formulations), would have looked to absorption enhancement agents to provide rapid onset and increase the systemic bioavailability with a lower dose.” Pet. 55 (citing, e.g., Ex. 1002 (Smyth Decl.) ¶¶ 541–42, 545–61; Ex. 1003 (Weinstein Decl.) ¶¶ 418–22). Petitioner contends that “Srisawat taught the addition of surfactant to increase uptake and absorption, and Maggio ’145 and Munjal directed POSAs to surfactants that are alkylglycosides, such as DDM.” *Id.* at 56 (citing Ex. 1002 (Smyth Decl.) ¶¶ 147–68, 532–33, 541–42, 545–61; Ex. 1003 (Weinstein Decl.) ¶¶ 157–76, 391–92, 406–20).

Similar to its obviousness argument for Potta, Maggio, and Munjal, Petitioner does not attempt to demonstrate that the additional limitation recited in claim 3, namely “a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis,” is expressly recited in the combination of Srisawat, Maggio, and Munjal. Instead, Petitioner asserts that “the additional limitation would have been an inherent result” of

intranasally administering Srisawat's 1.25 or 2.5 mg formulation with DDM. Pet. 57–58.

Neither the Petition nor Petitioner's cited declarant testimony (Ex. 1002 (Smyth Decl.) ¶¶ 576–81; Ex. 1003 (Weinstein Decl.) ¶¶ 433–38) provide a basis for concluding that “a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis” as recited in claim 3, would be the inherent result of the proposed method. Srisawat teaches that when the 1.25 and 2.5 mg formulations (without DDM) were administered intranasally, they failed to show systemic absorption of epinephrine as compared to saline. Ex. 1006 (Srisawat), Fig. 1. Thus, even if a person of ordinary skill in the art would have expected DDM to enhance the onset and bioavailability of epinephrine (*see, e.g.*, Ex. 1002 (Smyth Decl.) ¶ 563), the pertinent question is whether adding DDM to Srisawat's 1.25 and 2.5 mg formulations would *necessarily* result in the claimed plasma epinephrine concentration. We are unconvinced that Maggio and Munjal answer this question, because Petitioner has not demonstrated sufficiently that what applied to phenylephrine and sumatriptan would have also applied to the same extent to the claimed amounts of epinephrine. Indeed, the record reflects that “[e]ach drug is one particular case and, thus, the relationship between the drug characteristics, the strategies considered and the permeation rate is essential.” Ex. 1064 (Pires), 17.

To be sure, Srisawat teaches that administration of a 5 mg intranasal epinephrine formulations (which lacked DDM) resulted in epinephrine absorption, and that using a surfactant might promote the nasal absorption of epinephrine. *See* Ex. 1006 (Srisawat), Fig. 1, 5; Pet. 55. We are persuaded that if DDM were added to Srisawat's 5 mg formulation, it would be

possible to lower the dose from 5 mg and still likely achieve epinephrine absorption in the formulation recipients. *See, e.g.*, Ex. 1002 (Smyth Decl.) ¶¶ 550, 557, 559; Ex. 1020 (Munjal), 5–6, 9. What is missing on this record, however, is the extent to which Srisawat’s 5 mg dose could be lowered due to the addition of DDM, and still *necessarily* result in the claimed plasma epinephrine concentration. Or more to the point—nothing on this record indicates that if DDM were added to Srisawat’s 1.25 and 2.5 mg formulations, that they would *necessarily* result in the subject matter of claim 3. *See Par Pharm., Inc.*, 773 F.3d at 1194–95 (explaining that inherency is a “high standard,” requiring that Petitioner show “the limitation at issue [is] necessarily . . . present, or the natural result of the combination of elements explicitly disclosed by the prior art”).

As discussed above, this is not a case where a plasma epinephrine concentration is expected following administration of Srisawat’s 1.25 and 2.5 mg formulations with DDM. Rather, the record shows that some epinephrine intranasal dosages (e.g., Srisawat’s 5 mg dosage and Potta’s 6 mg dosage) resulted in epinephrine absorption, while others did not (e.g., Srisawat’s 0.3, 0.6, 1.25, and 2.5 mg dosages). *See, e.g.*, Ex. 1006 (Srisawat), 1, Fig. 1; Ex. 1004 (Potta), Example 8. Nothing on this record convinces us that Srisawat’s 1.25 and 2.5 mg formulations, even with the “boost” likely provided by DDM, would *necessarily* result in “a plasma epinephrine concentration that is efficacious for the treatment of anaphylaxis.” *See Par Pharm., Inc.*, 773 F.3d at 1194–95.

For at least the above reasons, we determine on the full trial record that Petitioner has not demonstrated by a preponderance of the evidence that claim 3 would have been obvious over Srisawat, Maggio, and Munjal.

Because claims 4–6 and 18–20 depend directly or indirectly from claim 3, and because Petitioner’s arguments for these claims do not rectify the deficiencies discussed above for claim 3, we similarly determine on the full trial record that Petitioner has not demonstrated by a preponderance of the evidence that claims 4–6 and 18–30 would have been obvious over Srisawat, Maggio, and Munjal.

III. PETITIONER’S MOTION TO EXCLUDE

Petitioner moves to exclude Exhibits 2015, 2019–2025, and 2031, as well as portions of Dr. Kaliner’s Declaration (Ex. 2013) and Dr. Wyse’s Declaration (Ex. 2014) that rely on those exhibits. Paper 37 (“Mot.”), 1; Paper 40 (“Reply”). Patent Owner opposes. Paper 29 (“Opp.”).

Petitioner argues that each of Exhibits 2015, 2019–2025, and 2031 are post-priority date evidence and Patent Owner improperly relies on them to indicate the state of the art today, not as of the priority date. Mot. 3–4, 5, 6. Petitioner further contends that Patent Owner relies on Exhibits 2019–2021 to argue unexpected results, based on an impermissible comparison of pre- and post-priority date data. *Id.* at 4. As such, Petitioner contends the exhibits are irrelevant and prejudicial and should be excluded under Federal Rules of Evidence (“FRE”) 401–403. As noted, Petitioner also seeks to exclude portions of Dr. Kaliner’s Declaration (Ex. 2013) and Dr. Wyse’s Declaration (Ex. 2014) that rely on those exhibits. Mot. 6–7; *see also* Exs. 1086, 1087 (highlighting portions of the declarations sought to be excluded).

Patent Owner contends that post-priority date evidence can be relevant in multiple contexts, including to demonstrate objective indicia of nonobviousness. Opp. 2. Patent Owner contends that its use of the exhibits

is permissible, and thus there is no basis to exclude the exhibits or the identified portions of the Kaliner and Wyse declarations. *Id.* at 2–9.

As the moving party, Petitioner bears the burden of establishing that it is entitled to the requested relief. 37 C.F.R. § 42.20(c). Having considered the arguments and evidence of record, we determine that Petitioner has not met its burden of proof. We agree with Patent Owner that “[n]o bright line rule prohibits the consideration of post-priority date evidence,” and that such evidence is permissible in certain contexts. Opp. 2–3. On this record we do not find that there is sufficient cause to exclude the challenged exhibits, because any consideration by the Board of these exhibits has taken into account their alleged post-priority date status. Thus, we do not find that maintaining the exhibits in this proceeding results in any unfair prejudice to Petitioner, or that the exhibits have confused the issues before us.

Accordingly, we deny Petitioner’s motion to exclude.

IV. CONCLUSION²¹

Based on the information presented, we conclude that that Petitioner has demonstrated by a preponderance of the evidence that claims 1, 2, 7–11, and 12–17 are unpatentable, but has not demonstrated that claims 3–6 and 18–20 are unpatentable.

²¹ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–18	102	Potta	1, 2, 7–17	3–6, 18
1–20	103	Potta, Maggio, Munjal	1, 2, 7–9, 11–17	3–6, 10, 18–20
1–20	103	Srisawat, Maggio, Munjal ²²		3–6, 18–20
Overall Outcome			1, 2, 7–17	3–6, 18–20

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims Petitioner has demonstrated by a preponderance of the evidence that claims 1, 2, and 7–17 of U.S. Patent 10,682,414 B2 are unpatentable, but has not demonstrated that claims 3–6 and 18–20 of this patent are unpatentable;

FURTHER ORDERED that Petitioner’s Motion to Exclude is *denied*; and

FURTHER ORDERED that, because this is a final written decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

²² We consider only claims 3–6 and 18–20 under this ground. We do not consider claims 1, 2, and 7–17 under this ground. *See supra* Section II.E.3.

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